Space Emanation Theory

Source, clock and interacting-field energy

Retained derivations from the source-history and quantum-phase investigation through the nonlinear inertia research.

Research synthesis - 6 November 2026

The central result is a common-energy construction for an evolving source. Within SET's retained local field and material realization, varying one complete source energy gives the internal clock rate, material velocity, material force and photon propagation. An independent source-law boundary test distinguishes field responses that energy equality alone allows. These are complementary results: one restricts the response; the other makes its interacting dynamics explicit.

Result to carry forward	What it contributes
Source-law discrimination	Actual supported sources test the Q boundary independently of their energy and force equations.
Complete source energy	Material, supports, electromagnetic stress and binding give compatible clock and exterior-source readings.
Electromagnetic evolution	Common local evolution selects a Maxwell branch within a stated vector class; electric work and magnetic turning have separate roles.
Quantum clock energy	A bounded clock operator and a positive energy complement survive noncommutation.
Interacting motion	The field-resolved energy generates clock, velocity, force and photon energy on an evolving slice.
Finite inertia response	Regular-source pulse calculations retain deformation, radiation and field feedback, with positive motion energy.
Source-to-light connection	Source reciprocity, photon redshift, source-volume cooling and actual field memory fit the same accounting.

The derivations below are arranged by dependency. Each new mathematical structure is named where it enters. Numerical examples are controlled calculations with stated inputs, not new observational fits. The older research remains the detailed record; this document identifies the results that should remain active in SET's next development.

Derivation map

Begin with capacity and source readings, then require one energy to account for their physical response. The stationary source and electromagnetic chapters establish concrete selection tests. The quantum chapter extends the clock-energy identity. The evolving-source chapters then derive force, exchange and inertia from the complete energy. The final optical chapter carries the same accounting to photons.

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1. Capacity, source and the readings we must keep distinct

In SET, a field's limited capacity for interaction means a limited capacity for physical motion. Mass continuously commits part of that capacity as gravitational burden. The remaining capacity is shared between translation and the local motion through which events unfold. A clock's journey and its ticks participate in the same physical limit.

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2, \qquad \beta = \frac{d\tau}{dT} \tag{1}$$

S is the burden in speed units. V_rel is the operational motion term per reference time T, and β measures internal progression relative to that reference. For a held observer in the zero-shift reading, write α=d t/dT and $S^2=c^2(1-\alpha^2)$. A local physical speed v then enters as V_rel=αv. The reference clock and the local observer's clock are different readings of the same process.

$$\alpha^2 = 1 - \frac{S^2}{c^2}, \qquad V_{\mathrm{rel}} = \alpha v, \qquad \beta = \alpha\sqrt{1 - v^2/c^2} \tag{2}$$

The source law is specified for a source domain. For a spherical reference volume $V_{ref}=4\pi R_0^3/3$, its equivalent forms are

$$\begin{aligned} Q_s^2 &= 24\pi G M_s V_{\mathrm{ref}} = 32\pi^2 G M_s R_0^3, \\ Q_s &= 4\pi\sqrt{2GM_sR_0^3}. \end{aligned} \tag{3}$$

The domain and its mass must be stated. A baryonic material domain uses its baryonic mass. A complete interacting source uses its complete mass-energy, including the field, supports and binding under the retained source-universality principle. A reference volume used to label a source is not automatically the proper volume integrated through a curved spatial reading.

$$\mathcal{H}_s = M_s c^2 = \frac{c^2}{24\pi G}\frac{Q_s^2}{V_{\mathrm{ref}}} \tag{4}$$

This quotient is the useful energy label. Changing the reference volume at unchanged complete energy changes Q_s but leaves Q_s^2/V_{ref} unchanged. Physically changing the source or its stored field state is a different operation.

The energy label alone does not select a field. At the actual outer material boundary of the admitted isolated neutral sources, with an empty exterior, SET also reads $Q_{boundary}=4\pi R^2 S(R)$. That value is calculated from the independently solved lapse and actual source radius, then compared with the energy-side source law. The supported-source chapter performs this test at a finite material boundary, rather than imposing a profile at every exterior radius.

A clock-normalized rate does not create a second source

$$Q_0\, dT = Q_m\, d\tau, \qquad Q_m = \frac{Q_0}{\beta}, \qquad Q_{\mathrm{held}} = \frac{Q_0}{\alpha} \tag{5}$$

Here Q_0 denotes the same reference-time rate in the two readings. The normalized rate is larger per slower clock second. It is not a second mass or an added reservoir. In a photon calculation, local observers supply the clock; a photon is not assigned a proper ticking clock. Source-volume evolution, rate conversion and ray propagation must retain their own definitions.

The elementary energy exchange is exact

At fixed probe rest mass m, multiply the invariant by m/2 and define $K_0=mV_{rel}^2/2$. Nothing has been added to obtain

$$K_0 = \frac{mc^2}{2}(1-\beta^2) - \frac{mS^2}{2}, \qquad dK_0 = -mS\,dS - mc^2\beta\,d\beta. \tag{6}$$

This is the direct capacity accounting the later work must respect. If both burden and translational motion increase, internal progression decreases sufficiently to balance them. K_0 is the quadratic motion reading; it is not required to remain constant along an orbit. The following chapters add the retained common-energy dynamics and test whether the same energy supplies the actual exchange.

How to read the derivation
Direct algebra means a rearrangement of explicitly stated equations. A retained completion means the previously adopted local field, canonical motion or wave structure used in that chapter. A selection result rules out alternatives within the class tested. This vocabulary identifies the mathematical starting point once, so the physical consequences can be followed without repeatedly interrupting the derivation.

[R1, §§1-3, 14-17; R12, §§1, 4-5.]

2. One common energy turns clock readings into a consistency test

The retained common-energy requirement is that a source's rest-energy response reads its physical clock. Let e_i be independent intrinsic source-energy coordinates and let X denote the remaining independent canonical state labels: positions, momenta, freely specified field data and supports. Constrained field components are re-solved after the source changes; holding X fixed does not freeze the entire physical field. Then

$$\beta_i = \frac{\partial \mathcal{H}}{\partial e_i}\Big|_X, \qquad \frac{\partial \beta_i}{\partial e_j} = \frac{\partial \beta_j}{\partial e_i} \tag{7}$$

The second identity follows from the equality of mixed derivatives. It says that changing two source energies in opposite orders cannot assign two energies to the same final complete state. Two histories may leave different field states; the requirement compares the same state, not merely the same material positions.

For the source-law variables, define σ_i=Q_i²/V_i and Σ=Q_total²/V_total. Intrinsic energies are e_i=c²σ_i/(24πG), and the complete energy has the same conversion factor. The requirement becomes

$$\frac{\partial \Sigma}{\partial \sigma_i} = \beta_i, \qquad \frac{\partial \beta_i}{\partial \sigma_j} = \frac{\partial \beta_j}{\partial \sigma_i},$$
$$d\mathcal{H}_{\text{source}} = \sum_i e_i \beta_i \left(2\frac{dQ_i}{Q_i} - \frac{dV_i}{V_i}\right). \tag{8}$$

Additional physical state changes carry their own work terms. Along an admitted family with 0≤β_i≤1 and no other independent work, a positive intrinsic energy increment gives 0≤dΣ≤Σ_i dσ_i. The initial field-energy baseline must be kept when integrating this inequality.

An immediate test for combining source readings

A proposed rule √Σ=√σ₁+√σ₂ gives ∂Σ/∂σ₁=1+√(σ₂/σ₁), exceeding one. It cannot represent these same held capacity-clock derivatives. The leading held-pair energy instead gives the reciprocal weak-field control

$$\Sigma = \sigma_1 + \sigma_2 - \frac{\sigma_1 \sigma_2}{24\pi c^2 r} \tag{9}$$

This is not a prohibition on adding all literal Q values. Under the separate bookkeeping assumptions M=ΣM_i and V=ΣV_i, with no binding correction or extra enclosing empty volume, the source law gives the exact identity

$$Q_{\text{union}}^2 - \left(\sum_i Q_i\right)^2 = 24\pi G \sum_{i<j} \left(\sqrt{M_i V_j} - \sqrt{M_j V_i}\right)^2 \tag{10}$$

Equal mean densities make literal Q additive. Unequal densities leave the positive difference shown. A complete interacting mass still requires its binding and field accounting. This distinction is necessary whenever SET combines several sources.

A supported two-source example supplies the clocks independently

Take two neutral concentric supported shells at radii R₁<R₂. Their proper material-plus-support energies are e₁,e₂. Put g=G/c⁴. The retained source jumps give the complete energy

$$E_{\text{in}} = e_1 - \frac{g e_1^2}{2R_1}, \qquad f_- = \sqrt{1 - \frac{2gE_{\text{in}}}{R_2}},$$
$$\mathcal{H} = E_{\text{in}} + e_2 f_- - \frac{g e_2^2}{2R_2}, \qquad f_+ = f_- - \frac{g e_2}{R_2}. \tag{11}$$

Differentiating at fixed radii gives the same lapses as the separately matched field:

$$\mathcal{H}_{e_1} = \alpha_1 = \frac{f_+}{f_-}\left(1 - \frac{g e_1}{R_1}\right), \qquad \mathcal{H}_{e_2} = \alpha_2 = f_+,$$
$$\mathcal{H}_{e_1 e_2} = -\frac{g}{R_2 f_-}\left(1 - \frac{g e_1}{R_1}\right) = \mathcal{H}_{e_2 e_1}. \tag{12}$$

The field response is already inside the derivative. It is not held artificially unchanged while a gravitating source changes energy. To allow the supports to relax, write e_i=b_i+U_support,i(R_i), with b_i an additive proper baseline energy and the support parameters fixed. Solving for stationary radii R*(b), the envelope H(b)=H(b,R*(b)) retains H_bi=α_i, while its second response includes the support relaxation:

$$\overline{\mathcal{H}}_{b_i b_j} = \mathcal{H}_{b_i b_j} - \mathcal{H}_{b_i R}(\mathcal{H}_{RR})^{-1}\mathcal{H}_{R b_j} \tag{13}$$

This is the first concrete bridge from the clock of a probe to the clock of a source that changes its own field. Later, the same question will be answered on an evolving matter-plus-photon slice, with no stationarity assumption. [R1, §§34-35, 40.]

3. Electromagnetic energy, source selection and directional motion

Electromagnetism tests more than whether a field carries energy. Its energy also carries directional stress, can change a probe's speed, and can turn its motion without changing that speed. The common-energy construction lets us follow those responses through the same source, clock and momentum account.

Retained framework and examined class
This section retains the selected static field energy, complete-energy source law and common canonical field completion. Stationary sources include their material supports. The Maxwell selection examines one local vector field with positive quadratic momentum and first-spatial-derivative energy in the stated unmixed canonical description. Signed electric charge and its coupling to matter are specified interaction data. The calculations identify what these premises select. [R1-R3]

Energy and physical stress enter the same source equation

Write the static ruler and clock as $q_{ij} = \psi^4\delta_{ij}$ and $\alpha = \chi/\psi$, with $dV = \psi^6 d^3x$. Δ is the reference Cartesian Laplacian. Proper energy density is ε; σ is spatial stress, so tr σ is the sum of the principal pressures. The retained field and material energies are

$$W_f = -\frac{c^4}{2\pi G}\int \nabla\psi\cdot\nabla\chi\, d^3x, \qquad W_m = \int \alpha\varepsilon\, dV$$
$$Q_s^2 = 24\pi G\frac{E_s}{c^2}V_{\mathrm{ref}} \tag{14}$$

E_s includes complete material, field, support and binding energy. Q_s is the gravitational source reading; electric charge will be e. At fixed canonical material data, changing the ruler performs work: $\delta U_m = -\tfrac{1}{2}\int \sigma^{ij}\delta q_{ij}\, dV$. Independent ψ and χ variations therefore give

$$\delta W_m = \int \left[\psi^5\varepsilon\,\delta\chi - \alpha\psi^5(\varepsilon + 2\mathrm{tr}\,\sigma)\delta\psi\right] d^3x \tag{15}$$

Stationarity of the same complete energy, followed by the lapse combination, gives

$$\Delta\psi = -\frac{2\pi G}{c^4}\psi^5\varepsilon, \qquad \Delta\chi = \frac{2\pi G}{c^4}\chi\psi^4(\varepsilon + 2\mathrm{tr}\,\sigma)$$
$$D^2\alpha = \frac{4\pi G}{c^4}\alpha(\varepsilon + \mathrm{tr}\,\sigma) \tag{16}$$

Here D uses the physical spatial ruler. Thus the stationary work current is $J_W = c^2 D\alpha$. Its complete-source flux encloses the full field tail at infinity:

$$D_i J_W^i = \frac{4\pi G}{c^2}\alpha(\varepsilon + \mathrm{tr}\,\sigma)$$
$$\lim_{r\to\infty}\oint_{S_r}\mathbf{J}_W \cdot d\mathbf{A} = \frac{4\pi G E_s}{c^2} = \frac{Q_s^2}{6V_{\mathrm{ref}}} \tag{17}$$

Stress is part of the source of clock-gradient work. Holding ε artificially fixed while varying the physical volume would remove that contribution. [R1, §6]

The charged source resolves the earlier stress discrepancy

A radial electric field has $\varepsilon = e^2/(32\pi^2\varepsilon_0 r^4)$ and principal stresses $(-\varepsilon,+\varepsilon,+\varepsilon)$. Consequently the conformal χ equation contains 3ε, while the physical lapse equation contains 2ε. Define $m = GE_s/c^4$, $d_e = Ge^2/(4\pi\varepsilon_0 c^4)$, $x = 1/\rho$ and areal radius $r = \rho\psi^2$. The spatial equation has the first integral

$$\psi_x^2 - \frac{d_e}{4\psi^2} = \frac{m^2 - d_e}{4}, \qquad \psi^2 = 1 + mx + \frac{m^2 - d_e}{4}x^2 \tag{18}$$

Keep the lapse coefficient b independent in $\alpha = 1 - b/\rho + O(\rho^{-2})$. The spatial boundary energy is $E_s = c^4 m/G$. Complete-source dilation, with the support stresses and decaying electric tail retained, gives the independent clock-current reading $E_s = c^4 b/G$. Equality of these complete energies selects b = m. The response is then

$$\alpha^2 = 1 - \frac{2m}{r} + \frac{d_e}{r^2}, \qquad B_r = \alpha^{-2}$$
$$S^2 = \frac{2GE_s}{c^2 r} - \frac{Ge^2}{4\pi\varepsilon_0 c^2 r^2} \tag{19}$$

B_r multiplies dr^2; it is unrelated to the magnetic field. The charged exterior follows through the retained energy and stress accounting. At a finite radius, its spatial-mass and work-current energy readings are different:

$$E_{\mathrm{out}} = \frac{e^2}{8\pi\varepsilon_0 r}, \qquad E_{\mathrm{spatial}} = E_s - E_{\mathrm{out}}, \qquad E_{\mathrm{work}} = E_s - 2E_{\mathrm{out}} \tag{20}$$

The additional contribution is supplied by electric stress. Both readings approach E_s at infinity. An electric field outside the material still belongs to the source account. In particular, the general static identity $d\ln(\alpha^2 B_r)/dr = (8\pi G/c^4) r B_r(\varepsilon + p_r)$ gives $\alpha^2 B_r = 1$ for the isolated radial Maxwell exterior because $\varepsilon + p_r = 0$. An externally clock-shifted vacuum gap can instead have a different constant, as the emitted-photon example below shows. [R1, §6]

A supported material source earns its clock response

Consider a charged material shell at areal radius R, enclosing a flat cavity. Let U_0 include all its proper material and support energy, and p_s be its tangential surface pressure. Integrating the spatial field equation through the shell gives $(dr/d\ell)_+ = 1 - GU_0/(Rc^4)$. Applying the exterior first integral gives

$$E_s = U_0 - \frac{GU_0^2}{2Rc^4} + \frac{e^2}{8\pi\varepsilon_0 R} \tag{21}$$

Material work is $dU_0 = -p_s\, d(4\pi R^2)$. Stationarity under a radius change fixes the support pressure. Its substitution into the independent lapse jump makes the clock and spatial factors obey the same boundary derivative and outer normalization. Thus

$$\begin{aligned} \alpha_s &= 1 - \frac{GU_0}{Rc^4}, \qquad \frac{\partial E_s}{\partial U_0}\Big|_{R,\,e} = \alpha_s \\ dE_s &= \alpha_s\left[dU_0 + p_s\, d(4\pi R^2)\right] + \frac{e}{4\pi\varepsilon_0 R}\, de \end{aligned} \tag{22}$$

The clock-energy equality is an output of the supported source calculation. A counterrotating massive-particle shell supplies an explicit support law $U_0 = \sqrt{(E^2_{\text{rest}} + c^2 J^2_{\text{abs}}/R^2)}$. Opposite orientations cancel net spin while providing pressure. The same energy gives

$$\beta_p = \alpha_s \frac{E_{\text{rest}}}{U_0}, \qquad \Omega = \alpha_s \frac{c^2 J_{\text{abs}}}{R^2 U_0}, \qquad S_s^2 + (R\Omega)^2 + c^2\beta_p^2 = c^2 \tag{23}$$

For $G=c=R=1$, $E_{\text{rest}} = \sqrt{58}/40$, $J_{\text{abs}} = \sqrt{6}/40$ and $\sqrt{d_e} = 1/10$, the squared capacity shares are exactly 9/25, 3/50 and 29/50. The fixed-material radial energy curvature is positive. This supplies an actual stationary material realization of source equilibrium, angular motion and internal progression. Physical angular action, rather than action divided by rest mass, is held fixed in the clock derivative. [R1, §7]

The independently checked Q boundary selects a coupling

The light speed alone leaves impedance undetermined. Within the reciprocal Maxwell family, allow a positive response $f(\psi,\chi)$, with permittivity $\varepsilon_0 f$ and permeability μ_0/f. Their product stays c^{-2}. At fixed canonical fluxes, let e_f and b_f be their nonnegative squared-flux coefficients. Then

$$h_E = \frac{\chi e_f}{\psi^3 f}, \qquad h_B = \frac{\chi b_f f}{\psi^3} \tag{24}$$

Use $X = \psi\partial\psi + \chi\partial\chi$ and $Y = -\psi\partial\psi + \chi\partial\chi$. Complete energy, stress and source dilation give an independently testable difference between clock-current energy E_C and complete energy E_s:

$$2(E_C - E_s) = \int (Y \ln f)(h_B - h_E)\, d^3x \tag{25}$$

Balanced waves are blind to this test. A supported neutral capacitor, made of two opposite-charge particle shells, has positive electric energy confined to their annulus. For a candidate satisfying $E_C = E_s$ for every such source, the identity gives a zero annular integral. Its pointwise consequence requires the actual source variation established below.

The second test compares two separately calculated source readings: $Q^2_{energy}=24\pi G(E_s/c^2)V_{ref}$, while $Q_{boundary}=4\pi R^2 S(R)$ uses the solved lapse and actual neutral outer-shell radius. Take $V_{ref}=4\pi R^3/3$. In the empty neutral exterior $\psi=1+a/\rho$, $\chi=1+b/\rho$ and $R=\rho\psi^2$. Complete-energy enclosure and the boundary algebra give

$$\mu_{\text{energy}} = \frac{GE_s}{c^4} = a - b, \qquad \mu_{\text{boundary}} = \frac{R(1-\alpha^2)}{2} = (a-b)\left(1 + \frac{a+b}{2\rho}\right) \tag{26}$$

Exact Q agreement means $\mu_{boundary}=\mu_{energy}$. At a finite positive boundary radius and nonzero positive source energy this selects $a+b=0$. The equality is tested at the actual material boundary; it is not imposed while solving the candidate source, nor at every arbitrary screen. [R1, §9]

The independent shell-work identity supplies the second response restriction. In $G=c=1$ units set $\xi=1/\rho$ and define the response momenta $p_\psi=-\chi'$ and $p_\chi=-\psi'$, with $K=-2p_\psi p_\chi$ and $X\cdot p=\psi p_\psi+\chi p_\chi$; primes mean ξ derivatives. Write the positive annular electric energy as $V_e=d_e\chi/(2\psi^3 f)$. Direct differentiation of the candidate radial equations gives

$$\frac{d}{d\xi}[X\cdot p - \xi(K - V_e)] = -\frac{1}{2}(X\ln f)V_e \tag{27}$$

The inner endpoint vanishes by the empty cavity and actual shell traction. The outer endpoint vanishes by the just-earned $a+b=0$ relation and outer shell traction. Thus the annular integral of $(X \ln f)V_e$ also vanishes. The two integral conditions can now be localized using the actual supported-source family.

For shell i, let M_i be its proper rest energy, J_i its total absolute angular action, U_i its complete proper particle energy, and $P_i=4\pi r_i^2 p_{s,i}$ its integrated tangential surface pressure; r_i is its areal radius and $p_{s,i}$ its surface pressure. The inner and outer shells have i=1 and 2, with charges +e and −e. At zero charge a strict two-shell equilibrium has a nonzero equilibrium Jacobian. A positive twice continuously differentiable f changes its equations by order e^2, so nearby equilibria persist for an open set of controls $z=(M_1,J_1,M_2,J_2,e)$, solved using the candidate's own field, jump and traction equations. Only for a candidate passing both source requirements for every preparation does $a+b=0$ place its data in the five-dimensional seed/endpoint chart $s=(\mu,\xi_2,p_\chi,\xi_1,e)$, with $\mu=a-b$ and $\xi_i=1/\rho_i$ for isotropic shell radii ρ_i. At the outer annular endpoint $\psi_2=1+\mu\xi_2/2$ and $\chi_2=1-\mu\xi_2/2$. The value p_χ there is a seed; shell traction fixes its other response momentum, with V_e evaluated at the same endpoint:

$$p_\psi = -\frac{\chi_2 p_\chi + \xi_2 V_e}{\psi_2 + 2\xi_2 p_\chi} \tag{28}$$

Let S map actual material controls z into that chart. Integrating the annulus and reading U_i,P_i from each shell jump reconstructs $M_i=\sqrt{[U_i(U_i-2P_i)]}$ and $J_i=r_i\sqrt{(2P_iU_i)}$, with $0<2P_i<U_i$. This defines a smooth left inverse G on actual equilibria. The Jacobian identity below makes DS full rank; both spaces have five dimensions, so the source image is open. Varying ξ_1 at fixed outer seed and nonzero e therefore changes an actual source while keeping its annular history fixed up to the endpoint. Differentiating the two zero integrals gives $(Y \ln f)V_e=(X \ln f)V_e=0$ there. Since $V_e>0$, both response derivatives vanish on the reached weak domain:

$$\mathcal{G}(\mathcal{S}(z)) = z, \qquad D\mathcal{G}\,D\mathcal{S} = I \tag{29}$$

$$Xf = Yf = 0 \quad \Longrightarrow \quad f_\psi = f_\chi = 0 \tag{30}$$

For the transparent subfamily $f=(\psi\chi)^n$, the endpoint difference is $-n$ times a strictly positive annular electric-energy integral. It vanishes only for $n=0$. The general result holds on each connected weak response domain reached by this supported-source family. The remaining constant is electromagnetic calibration.

The numerical controls solved each $f=\exp[\lambda(\psi\chi-1)]$ using its own static conformal asymptotic, field and material-force equations, with fixed particle populations, angular actions and electric charge. The Q boundary was withheld and evaluated afterward.

λ	Absolute fractional clock/energy error	Q^2 boundary / energy − 1
−1	6.7×10^{-15}	$+5.90490 \times 10^{-4}$
0	2.7×10^{-15}	$+1.8 \times 10^{-15}$
+1	1.6×10^{-15}	-5.72458×10^{-4}

A candidate static equilibrium can balance its own clock energy and forces yet fail the separately retained source law. The Q test therefore adds a genuine discriminator. This static source test is performed before the common-evolution restrictions developed next. These are theoretical equation controls. A constant parity-odd θ term remains locally a boundary term; electric capacitors cannot test it. The field-dependent logarithm $\theta=k \ln \psi+\theta_0$ survives a global static test but fails the retained local metric-only spatial-generator rule unless $k=0$. Source tests and local evolution contribute distinct restrictions.

Common evolution earns the Maxwell bulk

The retained common canonical completion uses the following gravitational normal-energy density in c=1 units. Here $q=\det q_{ij}$, P^{gij} is the canonical ruler momentum, $P_g=q_{ij}P^{gij}$ is its trace, and $R^{(3)}$ is the scalar curvature of the spatial ruler q_{ij}:

$$C_g = \frac{P_g^{ij}P_{ij}^{g} - P_g^2/2}{C_0\sqrt{q}} - C_0\sqrt{q}R^{(3)}, \qquad C_0 = \frac{1}{16\pi G} \tag{31}$$

Its common normal generators satisfy {C[N],C[M]}=D[v], with $v^i=q^{ij}(N\partial_j M-M\partial_j N)$. N is the common lapse, and D generates spatial relabeling. The canonical class contains one covector A_i and conjugate vector density D^i, constant isotropic quadratic coefficients, and an initially admitted regular potential $W(A^2)$. It has no derivatives of D^i, no gravity-vector momentum mixing or curvature-vector terms, and no extra normal or constitutive field. The test uses generic smooth local states and arbitrary common lapse. Begin with $T_{ij}=\nabla_i A_j$ and all three isotropic quadratic first-derivative terms:

$$C_{\nabla A}=\frac{\sqrt{q}}{2}\left[c_1 T_{ij}T^{ij}+c_2 T_{ij}T^{ji}+c_3(T_i{}^i)^2\right] \tag{32}$$

Its symmetric connection response is $Z^{ij}=(c_1+c_2)T^{(ij)}+c_3(\mathrm{tr}\,T)q^{ij}$. The mixed ruler-vector bracket contains $(K_{a\beta}Z^{a}{}_{\beta})A^i-2Z^{ia}K_{a\beta}A^b$. In this unmixed class arbitrary symmetric ruler velocity K and vector data require Z=0. Therefore

$$c_2=-c_1,\qquad c_3=0,\qquad C_{\nabla A}=\frac{c_1\sqrt{q}}{4}F_{ij}F^{ij},\qquad F_{ij}=\partial_i A_j-\partial_j A_i \tag{33}$$

The curl is selected. With $B^i=\varepsilon^{ijk}\partial_j A_k$ and canonical momentum density D^i, the remaining positive quadratic energy has coefficients a,b,d and determinant $D=ad-b^2$:

$$C_v=\frac{q_{ij}}{2\sqrt{q}}(aD^iD^j+2bD^iB^j+dB^iB^j),\qquad a>0,\quad \mathcal{D}>0 \tag{34}$$

Its true spatial generator is P_i-A_iG, with $P_i=F_{ij}D^j$ and $G=\partial_i D^i$. Direct normal composition differs from that generator by

$$\int v^i[(\mathcal{D}-1)P_i+A_i\mathcal{G}]d^3x \tag{35}$$

Independent generic states first select $D=1$. Preserving $A_iG=0$ then earns the source-free constraint $G=0$. If G were nonzero on an open set, A and its curl would vanish there; preservation would force D=0 there, contradicting nonzero divergence. No division by a vanishing field is used. The generated gauge constraint removes one canonical pair, leaving two transverse modes. Gauss preservation for arbitrary common lapse also removes the field-dependent part of a regular potential $W(A^2)$; its constant belongs to the unloaded-energy calibration. [R2, §§2-6]

Parameterize the remaining quadratic coefficients by f=1/a and $\vartheta=-b/a$. The determinant condition then gives $d=f+\vartheta^2/f$, and the energy becomes

$$\partial_i D^i=0,\qquad C_v=\frac{|D-\vartheta B|_q^2/f+f|B|_q^2}{2\sqrt{q}},\qquad f>0 \tag{36}$$

Eliminating D gives the Maxwell bulk action with constant positive calibration f and constant boundary angle ϑ. Composition closes modulo the generated internal gauge transformation. For the parity-even unit-normalized branch, D=−E in a local orthonormal reading; the physical flux below uses conventional electromagnetic signs. The two transverse frequencies are ω_T=Nc|k|, so local normal time dt=N dT gives local speed c and the retained photon Hamiltonian:

$$H_\gamma = \alpha c\sqrt{q^{ij}p_i p_j}, \qquad \frac{d\ell}{dT} = c\alpha, \qquad \alpha E_{\text{local}} = \text{constant} \quad \text{(stationary)} \tag{37}$$

A stronger conditional theorem removes the quadratic amplitude ansatz. In a regular first-derivative gauge class L(s,p), with s=(E²−B²)/2 and p=E·B in c=1 units, demand both polarizations follow this unchanged photon cone on every admitted EM background. The two independent linearized polarization equations force L_{ss}=L_{sp}=L_{pp}=0, hence L=a_0s+b_0p+d_0. Positive energy, unloaded calibration and the constant boundary term again leave Maxwell bulk. Exact all-background cone equality is stronger than a causal upper speed. [R3, §2]

Electric work and magnetic turning in source-clock variables

Differentiating capacity along a path with a ruler-compatible derivative gives

$$\mathbf{V}\cdot D_T\mathbf{V} = -\tfrac{1}{2}D_T(S^2) - c^2\beta D_T\beta \tag{38}$$

This fixes longitudinal exchange. A perpendicular response turns motion without changing those scalar shares. A local linear response doing zero work for every velocity is antisymmetric and therefore has the form Ω×V. Hamiltonian compatibility gives this directional structure an evolution rule. Set E_0=mc², where m here denotes the probe rest mass, and $\mathscr{E}$=√(E_0^2+c²$q^{ij}\pi_i\pi_j$), H_0=α$\mathscr{E}$ and β=∂H_0/∂E_0. In the local kinetic-momentum bracket class {x^i,π_j}=$\delta^i{}_j$ and {π_i,π_j}=eF_{ij}, Jacobi makes F momentum independent and spatially closed. Write $E_{T\,i}$=F_{i0} for the electric covector associated with reference time T; the local E used below is measured in proper normal time. Time-compatible evolution gives induction:

$$\nabla\cdot\mathbf{B} = 0, \qquad \partial_T\mathbf{B} + \nabla\times\mathbf{E}_T = 0$$
$$\frac{dH_0}{dT} = \frac{\partial H_0}{\partial T} + e\mathbf{E}_T\cdot\mathbf{V} \tag{39}$$

A local vector potential can be introduced after closure as a representation. The electric part changes energy; the magnetic part turns momentum. The explicit background-energy derivative remains present. [R3, §§12-14]

Use the selected local field energy u=(X²+Y²)/2, flux S_{EM}=cX×Y and momentum-flux tensor Π_{ij}=uδ_{ij}−X_iX_j−Y_iY_j, where X=√ε_0E and Y=B/√μ_0. Local energy continuity in the linear first-derivative class fixes the Maxwell curl equations up to a rotation of X,Y; induction removes that rotation. With the electric source Gauss constraint, complete momentum balance for local charge density ρ_e and current j_e gives

$$\partial_t\mathbf{g}_{\text{EM}} + \nabla\cdot\Pi = -\rho_e\mathbf{E} - \mathbf{j}_e\times\mathbf{B}, \qquad \mathbf{g}_{\text{EM}} = \mathbf{S}_{\text{EM}}/c^2 \tag{40}$$

Matter receives the opposite exchange. The transverse force is read from directional stress, not inferred from scalar power. Charge continuity preserves electric Gauss; the source constraint

supplies its initial value. The static reference-energy flux carries α^2 because energy and its rate each receive a clock conversion. Evolving geometry additionally carries stress work. [R3, §15]

For the admitted charged common energy $H=\alpha\mathscr{E}+e\Phi$, with scalar electric potential Φ and kinetic momentum $\pi_i=p_i-eA_i$, the mechanical energy is $H_{mech}=H-e\Phi=\alpha\mathscr{E}$. Keep local α and the probe rest-source reading Q_0 fixed. Then

$$Q_m = \frac{Q_0}{\beta}, \qquad H_{\mathrm{mech}} = \frac{\alpha^2 E_0 Q_m}{Q_0}, \qquad Q_m \mathbf{V} = \frac{Q_0}{m}\boldsymbol{\pi} \tag{41}$$

With local velocity $v=V/\alpha$, the exact work and directional equations become

$$\begin{aligned} \frac{dQ_m}{dT} &= \frac{Q_0}{E_0} e\mathbf{E}\cdot\mathbf{v} \\ \frac{D(Q_m\mathbf{V})}{dT} &= \frac{Q_0 \alpha e}{m}(\mathbf{E}+\mathbf{v}\times\mathbf{B}) \end{aligned} \tag{42}$$

Electric work changes the moving-clock reading. Ideal magnetic turning leaves Q_m and β unchanged while turning momentum. In a uniform magnetic field, let θ denote the signed turning angle and $\omega_T=d\theta/dT$ its reference-time angular frequency. The same clock gives

$$d\theta = -\frac{eB}{m}\,d\tau, \qquad |\omega_T| Q_m = \frac{|e|BQ_0}{m} \tag{43}$$

Equal-energy opposite-charge probes have the same scalar capacity and Q readings but turn oppositely. Signed source organization is therefore physical information carried by the admitted interaction. It is not an extra positive scalar occupancy. [R3, §16]

These results establish a connected branch: common energy and stress generate source and clock response; the independently checked Q boundary restricts a surviving coupling; common evolution selects Maxwell bulk in the stated local class; and the same probe energy organizes work and turning. Selection concerns physical dynamics up to equivalent variables. A canonical rewriting may introduce apparent mixing or momentum derivatives without changing any clock, source energy or Q. The gravitational source law remains universal while the electromagnetic interaction retains its signed source and probe information.

4. Orbital motion, internal clocks and source organization share one energy

The selected neutral spherical branch provides a useful anchor. Write $\mu=GM_s=Q_s^2/(32\pi^2R_0^3)$, $A=1-2\mu/(rc^2)$, and $E_0=mc^2$ for the probe. In the areal radial reading its retained Hamiltonian is

$$\mathcal{H} = \sqrt{A\left[E_0^2 + c^2\left(Ap_r^2 + L^2/r^2\right)\right]}, \qquad \mathcal{H}_{E_0} = \frac{AE_0}{\mathcal{H}} = \beta \tag{44}$$

E is the conserved reference energy, p_r the momentum conjugate to r, and L the probe angular momentum. The measurement radius is r; R_0 labels the specified source. For this solved exterior $S^2(r)=2\mu/r$. Consequently

$$S^2(r) = \frac{Q_s^2}{16\pi^2 R_0^3 r}, \qquad S(R_0) = \frac{Q_s}{4\pi R_0^2} \tag{45}$$

The second relation is the source-surface reading. Dividing the unchanged Q_s by $4\pi r^2$ at every exterior radius would define a different, r^{-2} speed profile. The two readings coincide at R_0, not at every r. Keeping this distinction protects all later source, energy and tidal calculations.

Differentiate the actual orbit

For a bound nonplunging orbit define the radial action I_r, reference radial period T_r, azimuthal angle Φ_r and elapsed proper time τ_r:

$$I_r = \frac{1}{\pi}\int_{r_p}^{r_a} p_r\, dr, \quad \Omega_r = \frac{2\pi}{T_r}, \quad \Omega_\varphi = \frac{\Phi_r}{T_r}, \quad \bar\beta = \frac{\tau_r}{T_r}, \qquad p_r = \frac{\sqrt{E^2 - A(E_0^2 + c^2L^2/r^2)}}{cA}. \tag{46}$$

The endpoint terms vanish because p_r vanishes at the regular turning points. Hamilton's equations then give $I_{r,E}=T_r/(2\pi)$, $I_{r,L}=-\Phi_r/(2\pi)$, and $I_{r,E_0}=-\tau_r/(2\pi)$. Inverting these derivatives gives

$$dE = \Omega_r\, dI_r + \Omega_\varphi\, dL + \bar\beta\, dE_0 + \langle \mathcal{H}_\mu \rangle\, d\mu \tag{47}$$

The exterior has one source length μ/c^2. Put $I=I_r+L$. Dimensional scaling and degree-one scaling of probe mass and actions give

$$E = mc^2 f\left(\frac{\mu m}{cI}, \frac{L}{I}\right), \qquad \mu E_\mu = mE_m - E, \qquad mE_m + I_r E_{I_r} + L E_L = E. \tag{48}$$

The orbital first law gives $mE_m=E_0\beta$, $E_{Ir}=\Omega_r$ and $E_L=\Omega_\varphi$. At fixed R_0, $\mu\propto Q_s^2$ gives $(Q_s/2)E_{Qs}=\mu E_\mu$. Combining these derivatives yields

$$\frac{Q_s}{2} E_{Q_s} = E_0\bar\beta - E = -\Omega_r I_r - \Omega_\varphi L \tag{49}$$

The source response, clock response and radial/angular motion are derivatives of the same energy. Source derivatives hold the actions, E_0 and R_0 fixed. A pure reference relabeling contributes through $2dQ_s/Q_s-3dR_0/R_0$, which vanishes at fixed physical mass.

Binding energy can be reconstructed from the clock family

$$\frac{d}{d\mu}\left(\mu\frac{E}{E_0}\right)=\bar{\beta},$$
$$\frac{E}{E_0}=\frac{1}{\mu}\int_0^{\mu}\bar{\beta}(s)\,ds=\frac{2}{Q_s^2}\int_0^{Q_s}Q'\bar{\beta}(Q')\,dQ'. \tag{50}$$

The lower boundary uses a regular bound family with $E\to E_0$ as $\mu\to 0$. The binding fraction is the corresponding average clock deficit across neighboring source strengths. This integration is not elapsed time along one orbit. The orbit dynamics supply the clock family being integrated.

For circular orbits put $u=\mu/(rc^2)$ and $k=\mu m/(Lc)$. The exact relations $k^2=u(1-3u)$, $\beta=\sqrt{(1-3u)}$ and $E/E_0=(1-2u)/\sqrt{(1-3u)}$ satisfy the reconstruction. At $u=1/6$ the squared capacity shares are $S^2/c^2=1/3$, $V_{rel}^2/c^2=1/6$ and $\beta^2=1/2$. The clock stays finite at this radial stability boundary.

Perihelion is a derivative of the same orbital energy

Expand the selected Hamiltonian through first post-Newtonian order, with $p^2=p_r^2+L^2/r^2$:

$$\mathcal{H}=mc^2+\frac{p^2}{2m}-\frac{m\mu}{r}$$
$$+\frac{1}{c^2}\left[-\frac{p^4}{8m^3}-\frac{\mu(3p_r^2+L^2/r^2)}{2mr}-\frac{m\mu^2}{2r^2}\right]+O(c^{-4}). \tag{51}$$

For a Kepler ellipse with semimajor axis a_K and eccentricity e, put $s_e=\sqrt{(1-e^2)}$, $I=m\sqrt{(\mu a_K)}$ and $L=Is_e$. With eccentric anomaly u_K, $r=a_K(1-e\cos u_K)$ and dt is proportional to $(1-e\cos u_K)du_K$. Time averaging gives the three required averages:

$$\langle p^4\rangle=\frac{m^4\mu^2}{a_K^2}\left(\frac{4}{s_e}-3\right),\qquad \langle r^{-2}\rangle=\frac{1}{a_K^2 s_e},$$
$$\left\langle\frac{3p_r^2+L^2/r^2}{r}\right\rangle=\frac{m^2\mu}{a_K^2}\left(\frac{4}{s_e}-3\right). \tag{52}$$

Their substitution gives the first-order energy at fixed actions:

$$E=mc^2-\frac{m^3\mu^2}{2I^2}+\frac{m^5\mu^4}{c^2}\left(\frac{15}{8I^4}-\frac{3}{I^3L}\right)+O(c^{-4}) \tag{53}$$

At fixed I, the L derivative gives the apsidal frequency. Divide it by the leading radial frequency $m^3\mu^2/I^3$, multiply by 2π for one radial cycle, and use the semilatus rectum $p=L^2/(m^2\mu)$. Then

$$\Delta\varphi=\frac{6\pi\mu}{pc^2}=\frac{3Q_s^2}{16\pi R_0^3 pc^2} \tag{54}$$

This is the leading advance per radial cycle. The new result carried forward here is the common source-clock-energy reconciliation of that dynamics, rather than a new observational perihelion fit. [R1, §§2-5.]

Physical organization and rotation contribute identifiable work

For a scale-covariant extended source with physical sizes b_a, the source identity becomes

$$\mu E_\mu + \sum_a b_a E_{b_a} = E_0\bar{\beta} - E \tag{55}$$

The b_a terms are the energy response to actual source organization. They survive a physical change of the source even if the reference Q label is rewritten. This gives a precise place for a derived nonspherical response to enter; the response family itself must still be supplied by its field/source calculation.

For the retained rotating exterior, define Q_J=8πGJ_s/c² and spin fraction χ=c³Q_J/(8πμ²). At fixed χ, Q_J scales as μ²; hence μE_μ at fixed χ equals μE_μ at fixed Q_J plus 2Q_JE_QJ. The same orbital scaling therefore gives the fixed-action identity

$$\frac{Q_s}{2} E_{Q_s}\Big|_{Q_J} + 2Q_J E_{Q_J}\Big|_{Q_s} = E_0\bar{\beta} - E = -\sum_i \Omega_i I_i \tag{56}$$

The spin term can reinforce or oppose the mass response. It cannot be discarded merely because the energy is expressed in Q. Similarly, a fixed-reference source Q_s and the rotating horizon-area reading Q_H=cA_H can change in opposite directions during rotational energy extraction. They use different domains. These are checks of the retained rotating completion, not a new derivation of that completion from a scalar budget. [R1, §§4, 10-13, 39.]

The tidal reading to retain
The neutral exterior shortcut K=12S⁴/(c⁴r⁴) uses the solved neutral spherical S(r). A charged region, an exterior radiation clock factor, or rotation changes the relationship. In the retained rotating branch a useful nonnegative tidal magnitude is √(K²+P²), where P is the dual curvature contraction. A clock burden, a source charge and a tidal scalar describe related but different quantities. None should be substituted for another without its solved field relation.

5. Quantum motion, phase and the same energy clock

The quantum continuation asks whether the same energy can still describe internal progression and motion when its operators do not commute. The answer is positive within a specified wave completion. A further magnetic calculation puts the source-clock motion directly into an exact operator identity. We then use the clock and common propagation ceiling to restrict the interactions allowed in stated local classes.

A positive clock response and its complementary energy

The retained common-energy rule is $\beta = \partial H/\partial E_0$, where E_0 is the proper internal rest-energy parameter. For the quantum comparison supply a static scalar wave Hamiltonian. The source, spatial measure, representation and boundary domain remain fixed during this derivative. Write [R1, §§25,32]

$$\begin{gathered} H(E_0) = \sqrt{K_{\mathrm{w}} + E_0^2 A}, \qquad A = \alpha^2, \\ K_{\mathrm{w}} \geq 0, \qquad 0 \leq A \leq \alpha_{\max}^2 I \leq I. \end{gathered} \tag{57}$$

K_w is a nonnegative self-adjoint wave operator with energy-squared units, independent of E_0. It differs from the earlier quadratic motion energy K_0. A and K_w need not commute. Define the clock response X and complementary energy Rq, then differentiate the squared energy:

$$\begin{gathered} X = \tfrac{\partial H}{\partial E_0}, \qquad R_q = H - E_0 X, \\ HX + XH = 2E_0 A, \qquad HR_q + R_q H = 2K_{\mathrm{w}}. \end{gathered} \tag{58}$$

With a positive spectral gap, the first Sylvester equation has the explicit positive solution

$$X = 2E_0 \int_0^\infty e^{-sH} A e^{-sH}\, ds \geq 0. \tag{59}$$

For $\alpha_{\max} > 0$, square-root monotonicity gives $H \geq E_0\sqrt{A} \geq E_0 A/\alpha_{\max}$. Apply the same positive Sylvester inverse to $\alpha_{\max} I - X$. It follows that $X \leq \alpha_{\max} I$. Independently, $H(E_0)/E_0 = \sqrt{(K_w/E_0^2 + A)}$ decreases as positive E_0 increases, giving $E_0 X \leq H$. For $\alpha_{\max} = 0$, $A = X = 0$ directly. Together these prove

$$0 \leq X \leq \alpha_{\max} I, \qquad H = E_0 X + R_q, \qquad R_q \geq 0. \tag{60}$$

The ceiling is the operator bound $\alpha_{\max} I$. It is not the generally false pointwise operator inequality $X \leq \alpha(x)$; spatially extended noncommuting states retain their wave response.

This positive split does not depend on treating the operators as ordinary numbers. The bound also has a finite form that survives a closing spectral gap. Set $H_1 = H(E_0)$, $H_2 = H(E_0 + \Delta)$, with $\Delta \geq 0$:

$$\begin{gathered} (H_1 + \Delta\alpha_{\max} I)^2 - H_2^2 \\ = 2\Delta(\alpha_{\max} H_1 - E_0 A) + \Delta^2(\alpha_{\max}^2 I - A) \geq 0, \\ 0 \leq H(E_0 + \Delta) - H(E_0) \leq \Delta\alpha_{\max} I. \end{gathered} \tag{61}$$

For $E_0 > 0$, $H^2 \geq E_0^2 A$ gives an integrable s^{-2} bound on the tail of the derivative kernel, extending the norm derivative to zero spectral infimum. Common zero modes are annihilated by A and K_w. At $E_0 = 0$ the finite inequality still holds, while a norm derivative need not exist. No rest clock is assigned to a photon through that endpoint.

Return the clock-energy share to the source law

Give the massive probe a source label Qp in one fixed reference volume Vp. It is distinct from the background source Qs that determines A and K_w. The source law and chain rule give

$$E_0 = \frac{c^2 Q_p^2}{24\pi G V_p}, \qquad E_0 X = \frac{Q_p}{2}\frac{\partial H}{\partial Q_p}, \\ H = \frac{Q_p}{2}\frac{\partial H}{\partial Q_p} + R_q. \tag{62}$$

If $K_w = \hbar^2 \mathcal{K}$ on a fixed domain, joint scaling of E_0 and $\hbar$ additionally gives $R_q = \hbar \partial H/\partial \hbar$. This is a mathematical scale derivative, not a change in Planck's constant. Its homogeneity premise must be retained.

$$H = \frac{Q_p}{2}\frac{\partial H}{\partial Q_p} + \hbar\frac{\partial H}{\partial \hbar}. \tag{63}$$

In the uniform commuting limit, the meaning is explicit:

$$H = \alpha\sqrt{E_0^2 + c^2 p^2}, \qquad X = \frac{\alpha E_0}{\sqrt{E_0^2 + c^2 p^2}}, \\ R_q = \frac{\alpha c^2 p^2}{\sqrt{E_0^2 + c^2 p^2}} = \mathbf{p}\cdot\mathbf{V}. \tag{64}$$

R_q is the momentum-scale contribution. It is not $H - \alpha E_0$; its slow-motion limit is twice that usual kinetic-energy difference. In a varying wave field it includes the supplied gradient response. For a static H, Heisenberg evolution gives the exact exchange

$$\{H, \dot{X}\} = 2E_0\dot{A} = -\frac{2E_0}{c^2}\frac{dS^2}{dT}, \\ \dot{R}_q = -E_0\dot{X}. \tag{65}$$

The two positive contributions exchange while their sum remains fixed. Here a dot is a Heisenberg derivative, $\dot{A} = (i/\hbar)[H,A]$, not an explicit change of the static background. These are common-domain or weak-form identities. In an explicitly changing background, $d\langle H\rangle/dT = \langle \partial H/\partial T\rangle$ instead accounts for work by the source. This is the connection to the complete-source calculation: the source must enter the same accounting when it is allowed to respond.

A finite clock consequence
For two supplied nonnegative rest-energy levels separated by Δ in one common motional preparation, the finite bound gives $|C(T)| \geq \cos[\Delta\alpha_{max} T/(2\hbar)]$ while the cosine argument is between 0 and π/2. It bounds how quickly the two motional branches become distinguishable. It supplies no universal maximum frequency. For compatible time-dependent first-order Hamiltonians the integral of α_{max} replaces $\alpha_{max} T$. [R1, §§30,32]

The exact quantum source-clock momentum

Supply a signed electric charge e and a quantum kinetic momentum Π. The magnetic response is encoded in the order of kinetic translations on their common domain: [R4]

$$\Pi_i = -i\hbar\partial_i - eA_i, \qquad [\Pi_i, \Pi_j] = ie\hbar\varepsilon_{ijk}B_k. \tag{66}$$

For a static magnetic field, zero electric potential and constant α > 0, retain H = α√(E₀² + c²Π²), with E₀ = mc² > 0. Its clock and the same source rate read by that clock are

$$\hat{\beta} = \alpha^2 E_0 H^{-1}, \qquad \hat{Q}_{\text{clock}} = Q_0\hat{\beta}^{-1} = \frac{Q_0 H}{\alpha^2 E_0}. \tag{67}$$

Use the actual position velocity $\hat{V}_i = (i/\hbar)[H, x_i]$. We cannot commute H through Π_i. The squared Hamiltonian gives the exact symmetric relation

$$\begin{aligned} \{H, \hat{V}_i\} &= \tfrac{i}{\hbar}[H^2, x_i] = 2\alpha^2 c^2 \Pi_i, \\ P_i^{(Q)} &\equiv \tfrac{1}{2}\{\hat{Q}_{\text{clock}}, \hat{V}_i\} = \tfrac{Q_0}{m}\Pi_i. \end{aligned} \tag{68}$$

Therefore the source-clock momentum itself exposes the magnetic commutator:

$$[P_i^{(Q)}, P_j^{(Q)}] = ie\hbar\frac{Q_0^2}{m^2}\varepsilon_{ijk}B_k. \tag{69}$$

$P^{(Q)}$ has units Q times velocity. The anticommutator is the exact quantum version of the classical product; it is not an unordered product or an inverse expectation value. The potentials, signed charge and quantum representation are stated inputs. With the corresponding temporal covariant derivative, Jacobi identities give the homogeneous Maxwell identities; the sourced equations require the common field energy and its material current.

Keep the source readings distinct
$Q_{source}^2 = 24\pi GMV$ is the finite-domain gravitational source. $Q_{clock} = Q_0/\beta$ expresses a specified rate in another clock. The historical microscopic normalization $Q_{micro} = \hbar^2/(m^2c) = c\lambda\!\!\!^{-}C^2$ uses a supplied Compton scale. Equal units do not make them equal: at $R_0 = \lambda\!\!\!^{-}C$, $Q_{source}/Q_{micro} = 4\pi\sqrt{2}\, m/m_P$, with $\lambda\!\!\!^{-}C = \hbar/(mc)$ and $m_P = \sqrt{(\hbar c/G)}$. Electric charge e is another quantity. [R4, §1]

Spin returns to a positive capacity equation

The supplied Pauli algebra gives $(\sigma\cdot\Pi)^2 = \Pi^2 - e\hbar\sigma\cdot B$. To test what the capacity relation actually selects, retain the positive family [R5]

$$\begin{aligned} K_\eta &= (1-\eta)\Pi^2 + \eta(\boldsymbol{\sigma}\cdot\boldsymbol{\Pi})^2, \qquad 0 \le \eta \le 1, \\ H_\eta &= \alpha\sqrt{E_0^2 + c^2 K_\eta}. \end{aligned} \tag{70}$$

Both constituent squares are positive. Their leading magnetic coefficient is η = g/2 in this family. The spin term commutes with position, so the same source-clock momentum identity applies. Substitution yields

$$\begin{aligned} &c^2(\alpha^2\hat{Q}_{\text{clock}}^{\,2} - Q_0^2) \\ &= (1-\eta)\sum_i (P_i^{(Q)})^2 + \eta(\boldsymbol{\sigma}\cdot\mathbf{P}^{(Q)})^2. \end{aligned} \tag{71}$$

Multiply on both sides by Q̂clock⁻¹, preserving order, and define

$$\begin{aligned} W_i = \sqrt{1-\eta}\, P_i^{(Q)}\hat{Q}_{\text{clock}}^{-1}, \qquad Z = \sqrt{\eta}\,(\boldsymbol{\sigma}\cdot\mathbf{P}^{(Q)})\hat{Q}_{\text{clock}}^{-1}, \\ S^2 I + \sum_i W_i^\dagger W_i + Z^\dagger Z + c^2\hat{\beta}^2 = c^2 I. \end{aligned} \tag{72}$$

This exact identity returns to the capacity form with positive terms. It holds for every admitted η, so the budget alone does not choose η = 1. The factors Wᵢ and Z are energy-defined contributions; they need not equal the position velocities. A wave current, its velocity fluctuations and its energy-flow speed are different specified readings.

The supplied single-square Pauli matter energy also supplies a source current. In its local nonrelativistic limit, the magnetic term is −Mmag·B with Mmag = (eℏ/2m)ψ†σψ. Varying the kinetic term with respect to A gives the convection current. Integrating the magnetic variation by parts gives its additional curl:

$$\begin{aligned} \mathbf{M}_{\text{mag}} &= \tfrac{e\hbar}{2m}\psi^\dagger\boldsymbol{\sigma}\psi, \\ \mathbf{j} &= \tfrac{e}{2m}\left[\psi^\dagger\boldsymbol{\Pi}\psi + (\boldsymbol{\Pi}\psi)^\dagger\psi\right] + \nabla\times\mathbf{M}_{\text{mag}}. \end{aligned} \tag{73}$$

Here ψ†ψ is particle-number density, or probability density for one particle. Mmag has magnetization units A/m and is distinct from the source mass M. The interface contribution is the surface current Mmag × n. This connects internal orientation to a current that can enter the common Maxwell energy. Charge, spinor representation and the Pauli coupling remain the specified matter input; material magnetic ordering and radiative corrections are not obtained by this variation. [R4]

For a supplied scalar Landau spectrum, the same clock gives a useful worked consequence. Rationalizing two adjacent levels gives

$$\begin{aligned} e_n^2 = E_0^2 + c^2p_z^2 + (2n+1)\hbar|eB|c^2, \qquad \beta_n = \tfrac{\alpha E_0}{e_n}, \\ \omega_{n+1,n}(Q_{\text{clock},n} + Q_{\text{clock},n+1}) = \tfrac{2Q_0|eB|}{m}. \end{aligned} \tag{74}$$

The transition phase frequency is set by the harmonic mean of the two clock factors. No frequency is fitted. The spectrum and charged coupling are supplied, and a transition probability or emission rate requires the interaction that produces it. [R1, §37; R4]

The clock restricts affine material response

Consider a supplied affine interaction in the common spatial reading, allowing its shifts a_i and u to depend on E_0. At fixed background, [R5]

$$H = \alpha\mathcal{E} + u, \qquad \mathcal{E}^2 = E_0^2 + c^2 q^{ij}(p_i - a_i)(p_j - a_j),$$
$$H_{E_0} = \frac{\alpha E_0}{\mathcal{E}} + u_{E_0} - V^i a_{i,E_0}. \tag{75}$$

The velocity already obeys capacity with the positive clock $\alpha E_0/\mathcal{E}$. If the actual internal clock is H_{E_0} and must obey the same relation for every momentum, then $u_{E_0} - V^i a_{i,E_0} = 0$. First set V = 0 and then choose independent velocity directions. The result is

$$u_{E_0} = 0, \qquad a_{i,E_0} = 0. \tag{76}$$

For $a_i = eA_i$ and $u = e\Phi$, the charge is fixed during this internal-energy variation within the same charge sector. The result does not assign its value or identify charges across different species. If the clock phase is itself assigned a charged gauge transformation, its operational rate must instead be covariant. That changes the clock identification; it cannot be silently combined with the original neutral phase proof.

Common causality restricts a field-dependent dipole

Keep a Maxwell baseline plus one local Pauli operator with a smooth coefficient f(I,J), where $I = (B^2 - E^2)/2$ and $J = E\cdot B$. Use local units $c = \varepsilon_0 = \mu_0 = 1$. At a point choose E and B parallel to z and an allowed rest spin density s along x with zero electric polarization. The interaction vanishes on this background, but its quadratic variation is [R5]

$$a\,\delta B_x \delta B_z + b\,\delta B_x \delta E_z,$$
$$a = s(Bf_I + Ef_J), \qquad b = s(-Ef_I + Bf_J). \tag{77}$$

For propagation along y the physical transverse characteristic polynomial is

$$P(v) = (v^2 - 1)(v^2 - 1 + 2bv) - a^2. \tag{78}$$

If $a \neq 0$, $P(1) = P(-1) = -a^2$ while P tends to positive infinity at both ends. There are real characteristic speeds above +1 and below −1. If a = 0 but $b \neq 0$, the additional roots $-b \pm \sqrt{(1+b^2)}$ again include $|v| > 1$. The common capacity ceiling therefore forces a = b = 0. The derivative map has determinant $B^2 + E^2$, so

$$f_I = f_J = 0. \tag{79}$$

The coefficient is constant on the tested connected field domain. The restriction is not caused by negative energy: the quadratic Hamiltonian is

$$h_2 = \frac{D_x^2}{2} + \frac{(D_z - bB_x)^2}{2} + \frac{B_x^2 + B_z^2}{2} - aB_xB_z. \tag{80}$$

Its leading Hessian minors are 1, 1, 1, $1 - a^2$, so arbitrarily small nonzero spin density supplies positive field-energy controls. The complete spinor principal block does not remove the excessive field speed. The result uses this Maxwell baseline, one specified operator and the stated freedom of spin orientation. Other compensating operators belong to a different class.

A constant additional dipole d survives this local test. Actual coupled Maxwell-Dirac-Pauli controls starting with B = D = 0 generate different fields for d = .03 and .06 while preserving their positive clock-capacity sector and common energy. At the same final time their physical field energies are 4.27973×10^{-5} and 1.71255×10^{-4} in the declared local units. These are nonlinear local matter-field controls with a classical commuting spinor whose preparation contains both free-energy bands; gravity is not evolved. Their Q readout normalizes the same energy and is not an independent gravitational-boundary check.

What is retained
The exact quantum energy split and source-clock momentum are constructive bridges. The affine-clock and variable-dipole calculations are actual selectors within stated classes. Charge, quantum representation and the remaining constant interaction are not uniquely fixed by these results. The stronger all-field dipole clock gate is kept in the research ledger because it additionally requires continuation through a spectral gap closure and arbitrarily strong admissible sources. It does not erase the minimal charged magnetic moment.

Independent synthesis checks reproduce 16 exact symbolic identities, 24 noncommuting positive-operator controls, five positive spin-capacity families and 30 spinor identities. The maximum new operator residuals are below 2×10^{-14}. These are algebra and implementation checks; the general statements follow from the displayed proofs, not numerical fitting.

6. One complete source energy for clocks, motion and inertia

The central result of this part is that material clock progression, material motion and light arise as different responses of one complete source energy. The field is re-solved in each variation. The same calculation supplies the actual material momentum equation and a positive energy contribution from motion. It applies to moving, radiating spherical sources within the retained common-clock field and material completion. [R12]

The preceding tests explain why the complete account matters. A scalar memory control produced a causal acceleration reaction and no uniform-motion drag, but only a gravitational self-field correction. The selected-field calculation then included the actuator, support stresses and outgoing tensor waves: the retarded work matched the emitted energy, while the complete binding correction to inertia was the binding energy divided by c^2. [R6, R7]

Complete inertia, before assigning a body dispersion

For the retained isolated field system, differentiable boundary generators supply the boost flow of the full energy and momentum, with rapidity ζ. Starting from a rest state and integrating that flow gives: [R8]

$$\begin{gathered}\frac{dE}{d\zeta} = cP, \qquad \frac{dP}{d\zeta} = \frac{E}{c} \\ E = E_{\text{rest}}\cosh\zeta, \qquad P = \frac{E_{\text{rest}}}{c}\sinh\zeta \\ E^2 - c^2P^2 = E_{\text{rest}}^2, \qquad M_{\text{inertia}} = \frac{E_{\text{rest}}}{c^2} = \frac{Q_\Omega^2}{24\pi G V_\Omega}\end{gathered} \tag{81}$$

These are charges of the material and its field together. The rest energy already includes internal motion, support and binding. The microscopic material rest-energy parameters remain inputs; the result determines the complete bound system's inertia. The source-law volume belongs to the specified source domain; it is not an arbitrary exterior reading sphere. The argument uses isolated asymptotic boundary conditions and the retained translation/boost generators.

A separate regular-source calculation obtains the same inertia from the actual quadratic material and field motion terms. The local shift is solved, and their sum reduces to the static source mass at the boundary. Five supported sources agree with the independent static-energy integral to a maximum relative residual of 2.42×10^{-12}. This result keeps all static binding orders and is quadratic in collective velocity. [R9]

$$dE - \mathbf{v}\cdot d\mathbf{P} = \beta_{\text{bulk}}\, dE_{\text{rest}}, \qquad \beta_{\text{bulk}} = \gamma^{-1} \tag{82}$$

The last identity keeps the energy spent by an accelerating source. Its bulk translation factor is different from a material clock inside the source, which also participates in the source's own burden.

The moving source and its field

Use G = c = 1 in the following derivation. In polar areal coordinates, r is the areal radius, N is the reference-clock factor and a is the radial spatial coefficient. Let v be local radial material speed and W its Lorentz factor:

$$\begin{gathered}ds^2 = -N^2dT^2 + a^2dr^2 + r^2d\Omega^2, \qquad a = (1-2m/r)^{-1/2} \\ W = (1-v^2)^{-1/2}, \qquad \frac{dr}{dT} = \frac{Nv}{a}, \qquad \beta = \frac{N}{W} \\ S^2 = 1 - N^2, \qquad V_{\text{rel}} = Nv, \qquad S^2 + V_{\text{rel}}^2 + \beta^2 = 1\end{gathered} \tag{83}$$

The capacity identity is the reading of the same state. Its dynamical content is tested below by deriving the clock, velocity and force from the complete energy. E, j and P_r denote the total normal-frame energy, radial momentum and radial stress of material plus photons. The field equations are:

$$\begin{aligned} m_r &= 4\pi r^2 E, \qquad \frac{N_r}{N} = a^2\left(\frac{m}{r^2} + 4\pi r P_r\right) \\ m_T &= -4\pi r^2 \frac{N}{a} j, \qquad a_T = -4\pi r N a^2 j \end{aligned} \tag{84}$$

M = m(∞) is the complete energy in these units. The coordinate volume element is $dV_c = 4\pi r^2 dr$, and proper spatial volume is $dV = a\, dV_c$. The spherical reduction retains the field's changing binding. Its angular extrinsic curvature vanishes, so the reduced gravitational kinetic scalar and bulk symplectic term vanish after the constraints are imposed; m, a and N still evolve. [R12]

A number-conserving fluid has proper baryon density n, proper energy μ per baryon and internal energy density e(n). Its rest-frame energy density ε, pressure p and specific enthalpy h obey the same material law:

$$\varepsilon = \mu n + e(n), \qquad p = n e'(n) - e(n), \qquad h = \frac{\varepsilon + p}{n} = \mu + e'(n) \tag{85}$$

The pulse uses $e(n) = Kn^2$ and $\mu = m_b + z$, where z is fuel energy per baryon. Pressure supplies the source's regular support. The analytical derivatives extend to the causal class h > 0 and $0 \le c_s^2 = ne''/h \le 1$. The material and engine are specified preparations; no trajectory or complete-body inertia is fitted.

$$\begin{aligned} E_m &= (\varepsilon + p)W^2 - p, \qquad j_m = (\varepsilon + p)W^2 v, \\ P_{r,m} &= (\varepsilon + p)W^2 v^2 + p. \end{aligned} \tag{86}$$

Canonical source variation gives the clock weight

Use baryon inventory and canonical momentum per baryon as the independent material data. A μ variation fixes that inventory, positions and Π; a Π variation fixes inventory, positions and μ:

$$\begin{aligned} d &= anW, \qquad dN_b = 4\pi r^2 d\, dr, \qquad \Pi = ahWv \\ \Pi &= a\mu W v + 2Kd\, v, \qquad E_m = \frac{\mu dW}{a} + \frac{Kd^2}{a^2}(1 + v^2) \end{aligned} \tag{87}$$

The second line displays the actual quadratic material law. Differentiating at fixed canonical momentum includes the implicit change of local velocity when the radial field changes. The resulting local derivatives are:

$$\begin{aligned} E_{m,\mu} &= n, \qquad E_{m,\Pi} = \frac{dv}{a^2}, \qquad E_{m,d} = \frac{hW}{a} \\ E_{m,\ln a} &= -(E_m + P_{r,m}), \qquad \Pi_v = ahW^3(1 - c_s^2 v^2) > 0 \end{aligned} \tag{88}$$

For example, $E_{m,v} = (dv/a^2)\Pi_v$ gives the momentum derivative. At fixed Π, the μ-induced velocity change cancels the excess W factor, leaving $E_{m,\mu} = n$. The same identities hold for the general causal material class above.

Photons must also be held at a canonical phase state. For the spherical distribution $F(r, p_r, \ell)$, with ℓ the angular momentum squared, both ray energy and proper-volume conversion depend on a:

$$\begin{gathered}\varepsilon_\gamma = \sqrt{p_r^2/a^2 + \ell/r^2}\,, \qquad E_\gamma = \tfrac{\pi}{ar^2}\int F\varepsilon_\gamma\, dp_r\, d\ell \\ P_{r,\gamma} = \tfrac{\pi}{ar^2}\int F\tfrac{p_r^2}{a^2\varepsilon_\gamma}\, dp_r\, d\ell, \\ d\Gamma = 4\pi^2 dr\, dp_r\, d\ell, \qquad E_{\gamma,\ln a} = -(E_\gamma + P_{r,\gamma})\end{gathered} \tag{89}$$

Holding local intensity fixed would instead vary a different state. Combining the material and photon derivatives with $\delta \ln a = a^2\delta m/r$ gives the complete source-constraint variation:

$$\begin{gathered}(\delta m)_r + 4\pi r a^2(E + P_r)\delta m = 4\pi r^2 \delta E_{\text{explicit}} \\ [\ln(Na)]_r = 4\pi r a^2(E + P_r)\end{gathered} \tag{90}$$

The second line follows from the same field equations. Therefore Na is exactly the integrating factor. Regularity sets $\delta m(0) = 0$, and exterior vacuum normalization gives $Na \to 1$. Integration then yields:

$$\begin{gathered}\delta M = \int Na\, \delta E_{\text{explicit}}\, dV_c \\ \delta M = \int (\beta\, \delta\mu + \tfrac{Nv}{a}\delta\Pi) dN_b + \int H_\gamma \delta F\, d\Gamma \\ \tfrac{\delta M}{\delta\mu(N_b)} = \beta, \qquad \tfrac{\delta M}{\delta\Pi(N_b)} = \tfrac{dr}{dT}, \qquad \tfrac{\delta M}{\delta F} = H_\gamma = N\varepsilon_\gamma\end{gathered} \tag{91}$$

One complete source energy generates the material clock, material velocity and photon Hamiltonian. This is a first variation of the exact nonlinear constraint on an evolving mixed state. Neither a stationary source nor a weak background was required. [R12]

Read the same dynamical energy directly in Q

Restore physical energy units and apply the retained complete-source correspondence to the boundary mass just calculated. The chosen domain refers to the complete material-plus-photon system assigned that energy, not a smaller remnant that excludes escaped radiation. For the specified source-domain convention define:

$$\mathcal{E}_Q \equiv \tfrac{c^2 Q_\Omega^2}{24\pi G V_\Omega} = M_{\text{boundary}} c^2 = \mathcal{H}_{\text{complete}} \tag{92}$$

Let e_0 be rest-and-fuel energy per baryon, p_i canonical momentum per baryon, and F the canonical photon state. The already-derived variation now has a direct source-law reading:

$$\delta\mathcal{E}_Q = \int \left(\beta\,\delta e_0 + \frac{dx^i}{dT}\delta p_i\right) dN_b + \int H_\gamma \delta F\, d\Gamma,$$
$$\frac{\delta\mathcal{E}_Q}{\delta e_0(N_b)} = \beta, \qquad \frac{\delta\mathcal{E}_Q}{\delta p_i(N_b)} = \frac{dx^i}{dT}, \qquad \frac{\delta\mathcal{E}_Q}{\delta F} = H_\gamma \tag{93}$$

The functional derivatives use the displayed baryon and phase measures; positions are fixed in this line and varied next. In the spherical calculation the material index is radial. The isolated asymptotic boundary supplies M_boundary and the normalization of T. This is the same selected complete energy expressed through Q, not a new independent dynamics selection. If the chosen source volume changes, keep both terms:

$$\delta\mathcal{E}_Q = \frac{c^2}{24\pi G}\left(\frac{2Q_\Omega \delta Q_\Omega}{V_\Omega} - \frac{Q_\Omega^2 \delta V_\Omega}{V_\Omega^2}\right) \tag{94}$$

These derivatives also return the full source energy to the original capacity relation. Since the actual radial motion has V_rel=a dr/dT, the same re-solved state obeys

$$S^2 + a^2\left[\frac{\delta\mathcal{E}_Q}{\delta p_r(N_b)}\right]^2 + c^2\left[\frac{\delta\mathcal{E}_Q}{\delta e_0(N_b)}\right]^2 = c^2. \tag{95}$$

Here p_r is material canonical momentum per baryon, the functional measure is dN_b, and E_Q contains the complete ratio Q^2/V. The variations compare neighboring source states with the field re-solved; they are not the time derivative of one conserved energy. This is an exact capacity identity satisfied by the retained interacting dynamics. It verifies their common origin in this completion; the scalar identity by itself does not uniquely select the functional.

The position variation supplies the actual force

Return to G=c=1. Move material labels radially by ξ while keeping each label's μ and Π attached to it. Baryon conservation and advection determine the variations:

$$\delta d = -r^{-2}(r^2 d\xi)_r, \qquad \delta\mu = -\xi\mu_r, \qquad \delta\Pi = -\xi\Pi_r \tag{96}$$

Using the same energy derivatives and integrating by parts gives:

$$\delta M = \int \xi\left[(NhW)_r - \frac{Nv}{a}\Pi_r - \beta\mu_r\right] dN_b \tag{97}$$

The boundary term vanishes at the regular center and physical vacuum surface, or for a compact displacement. Each label's Hamilton equation is the negative position derivative:

$$\Pi_T|_{\text{label}} = -(NhW)_r + \frac{Nv}{a}\Pi_r + \beta\mu_r, \qquad \Pi_T|_{\text{label}} = \Pi_T + \frac{Nv}{a}\Pi_r \tag{98}$$

The transport terms cancel when the equation is written at fixed areal radius:

$$\Pi_T + (NhW)_r = \beta\mu_r \tag{99}$$

This is the conservative radial Euler equation, including pressure and composition gradients. Independent covariant differentiation and the conservative equation used by the numerical solver give the same law. The energy has therefore selected actual momentum exchange within this retained system, not just evaluated a clock along a supplied trajectory.

Its instantaneous rest limit makes the force explicit. With v=0 and no emission, h_r=μ_r+(∂_r p)/n gives Π_T=−hN_r−N(∂_r p)/n=a h v_T. Therefore, in c=1 units,

$$\begin{aligned} \frac{dv}{d\tau} &= -\tfrac{1}{a}\partial_r \ln N - \frac{\partial_r p}{a(\varepsilon+p)}, \\ \partial_r p &= -(\varepsilon+p)\partial_r \ln N \quad \text{(equilibrium)}. \end{aligned} \tag{100}$$

The velocity is measured relative to the local held frame; ∂_r p is the pressure gradient, distinct from canonical photon momentum p_r. For weak gravity, a≈1 and N≈1+Φ, the same law gives gravitational acceleration −∂_rΦ plus the pressure acceleration.

The declared fuel engine has dz/dτ = −Γz and local emitted power L = nΓz. For isotropic rest-frame emission, its source-only tangent at fixed instantaneous geometry, n and v is:

$$\begin{aligned} \mu_T = -\tfrac{N\Gamma z}{W}, \qquad \Pi_T|_{\text{emit}} &= -aN\Gamma z\, v \\ \Pi_T + (NhW)_r &= \beta\mu_r - aN\Gamma z\, v \end{aligned} \tag{101}$$

Clock and momentum derivatives now give the same complete energy transferred into the newborn photons:

$$\dot{M}_m\big|_{\text{emit}} = -\int N^2 L W\, dV, \qquad \dot{M}_\gamma\big|_{\text{emit}} = +\int N^2 L W\, dV \tag{102}$$

The cancellation is exact. The momentum term supplies the portion a clock-only calculation would omit when material is moving. The photons subsequently follow their Hamiltonian on the evolving field, with dHγ/dT = ∂_THγ. A stationary photon energy cannot be retained in a changing field by assumption. [R11, R12]

Positive motion energy with binding included

Keep the material positions, full canonical baryon-density profile d(r), composition and the entire canonical photon state fixed. Scale $\Pi_\lambda = \lambda\Pi_{\text{final}}$, for $0 \le \lambda \le 1$, and re-solve the field. The same derivative theorem gives:

$$\frac{dM}{d\lambda} = \int \tfrac{N_\lambda v_\lambda}{a_\lambda}\Pi_{\text{final}}\, dN_b \ge 0 \tag{103}$$

Positive enthalpy makes momentum and velocity have the same sign. Thus actual radial material motion has a positive complete energy cost, including field readjustment and photons still overlapping the source. The comparison stays on the regular untrapped branch. λ labels states; it is not an assumed physical history.

$$E^{(2)}_{\text{motion}} = \tfrac{1}{2}\int N_0(\varepsilon_0 + p_0)\tfrac{v^2_{\text{physical}}}{c^2}\, dV_0 \tag{104}$$

This is the quadratic limit for an isolated momentarily static companion. Its clock and enthalpy weights follow from the source constraint. The companion has the same inventory and composition at zero radial momentum; it need not be an equilibrium. Motion and deformation can consequently be separated without fitting an inertial coefficient.

$$E_{\mathrm{motion}} = \frac{c^2}{24\pi G}\left(\frac{Q^2_{\mathrm{moving}}}{V_{\mathrm{moving}}} - \frac{Q^2_{\mathrm{rest}}}{V_{\mathrm{rest}}}\right) \tag{105}$$

This is the source-law reading of the derived energy difference. Each volume retains its stated source convention: the areal reference volume $4\pi R_0^3/3$ is distinct from the proper-volume integral. It can stay fixed when the outer areal radius is fixed, even if the proper spatial measure changes.

The local response also resolves the direction of an impulse

In an orthonormal local frame with c=1, momentum per baryon is p_i=hWv_i. Hold the local ruler, composition and normal-frame baryon density D=nW fixed. Then $\partial n/\partial v_j=-nW^2 v_j$. Differentiating the same material momentum gives the exact directional response:

$$\begin{aligned} \mathcal{I}_{ij} \equiv \frac{\partial p_i}{\partial v_j} &= hW\delta_{ij} + hW^3(1-c_s^2)v_i v_j, \\ \mathcal{I}_\perp = hW > 0, &\qquad \mathcal{I}_\parallel = hW^3(1-c_s^2 v^2) > 0. \end{aligned} \tag{106}$$

The momentum response along the motion differs from the transverse response, and the material sound speed controls its compression contribution. This is a local invertibility and inertia result, not the Hessian of the complete constrained source energy. Converting radial local momentum to Π=a p_local,r gives Π_v=a I_parallel. It supplies a concrete ingredient for the next directed finite-source calculation without assigning a global recoil in advance. [R12, §7.]

Apply that response to a specified photon impulse

The same local tensor turns the declared engine into an actual velocity change. In c=1 units use the rest-frame emission pattern L(1+b cos θ)/(4π), with |b|≤1, along the same local axis as v. Dots in this paragraph mean only the emission contribution to the reference-time derivative, with D=nW and instantaneous N,a fixed. Boosting the emitted four-momentum gives

$$J^{\hat{0}} = LW(1+bv/3), \qquad J^{\hat{r}} = LW(v+b/3), \qquad L = n\Gamma z. \tag{107}$$

Let j_m denote normal material momentum density, distinct from the burden S. Its energy and momentum obey E_m=−NJ⁰ and j_m=−NJʳ. Using (μ,v) as independent variables at fixed D, the direct derivatives are E_m,μ=DW, j_m,μ=DWv, E_m,v=Dv I_parallel and j_m,v=D I_parallel. Here the μ derivative fixes v; the earlier derivative E_m,μ=n instead fixed canonical Π. Subtracting v times momentum conservation from energy conservation first fixes the fuel response. Substitution back into momentum conservation gives

$$\dot{\mu} = -\frac{N\Gamma z}{W}, \qquad \dot{\Pi} = -aN\Gamma z(v + b/3),$$
$$\dot{v} = -\frac{N\Gamma z\, b}{3hW^3(1 - c_s^2 v^2)} = -\frac{\Gamma z\, b\, \beta^3}{3h[N^2(1 - c_s^2) + c_s^2 \beta^2]}. \tag{108}$$

The second equality uses β=N/W from the invariant. At v=0 the reference-time response is −NΓzb/(3h); division by the local clock rate N gives the local response. Compression is included because n=D/W changes with v. The clock-plus-momentum energy loss is −N²Γz(1+bv/3) per baryon, exactly opposite to the newborn photons. This local source response joins the pressure, transport and changing-field terms in the full evolution. The β³ numerator alone is not a universal suppression law: its denominator retains N, h and the sound speed. [R12, §7.]

The complete energy has a positive clock-linked partition

The local material algebra also gives $E_m - \Pi E_{m,\Pi} = \varepsilon = \mu n + e(n)$. Inserting the complete-energy derivatives yields:

$$M = \int \mu\beta\, dN_b + \int \Pi \tfrac{Nv}{a}\, dN_b + \int F H_\gamma\, d\Gamma + \mathcal{R}$$
$$\mathcal{R} = \int [(1 - Na)E + Na\, e(n)] dV_c \geq 0 \tag{109}$$

Nonnegative E, E + Pᵣ and e(n), together with [ln(Na)]ᵣ ≥ 0 and exterior Na → 1, give 0 < Na ≤ 1. Momentum and photon terms are also nonnegative. In physical energy units:

$$\mathcal{H}_{\text{complete}} = \int e_0 \beta\, dN_b + \mathcal{E}_{\text{complement}}, \qquad \mathcal{E}_{\text{complement}} \geq 0 \tag{110}$$

e₀ is the carried rest-mass-and-fuel energy per baryon; compression energy remains in e(n). This is a partition of one energy, not energy consumed by ticks. The momentum term is p·v, not conventional kinetic energy with a one-half factor, and the remainder is not a separate localized gravitational energy density.

For e(n) = Kn² = p, the remainder has an exact Euler form. Restore G temporarily inside a² = (1 − 2Gm/r)⁻¹ and differentiate at fixed canonical source state:

$$KM_K = \int Na\, p\, dV_c, \qquad -GM_G = \int (1 - Na) E\, dV_c$$
$$\mathcal{R} = KM_K - GM_G \tag{111}$$

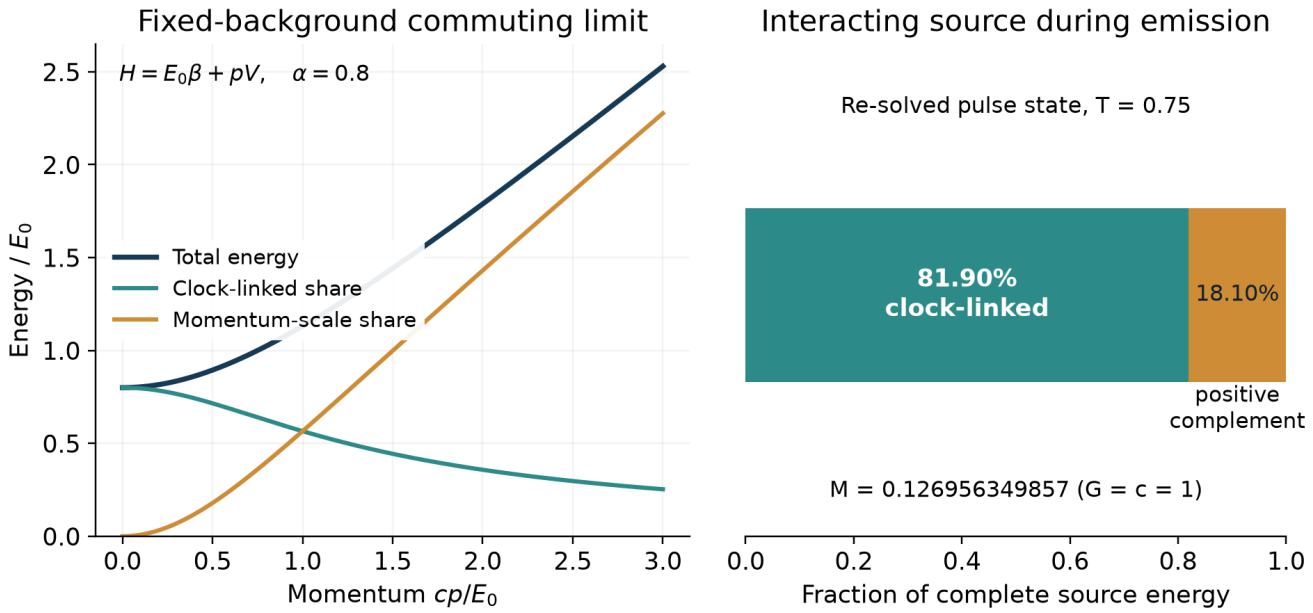
For the second identity, put y = ∂m/∂G and f = [ln(Na)]ᵣ. Differentiating the source constraint gives $y_r = -f(y + m/G)$. Hence Z = m + Gy obeys $Z_r + fZ = m_r$, whose integrating factor gives M + GM_G = ∫NaE dV_c. The first identity follows from $E_{m,K} = n^2$. These are sensitivities, not physical changes of G or the material law.

An auxiliary scaling of μ, Π, K and F by λ, accompanied by G → G/λ, keeps geometry and capacity readings fixed and sends M → λM. For the same source domain it also keeps $Q^2 = 24\pi GMV$ fixed. This checks compatibility with the source law without adding a new dynamical premise.

Two limits clarify the partition. Flat dust gives the classical scalar clock split. Static isolated polytropic equilibrium gives a virial relation from the source and field equations:

$$H = \sqrt{e_0^2 + c^2 p^2} = e_0\beta + \mathbf{p}\cdot\mathbf{v}$$
$$M = \int Na(\varepsilon + 3p)dV_c, \qquad -GM_G = 3KM_K \tag{112}$$
$$M - \int \mu\beta\, dN_b = 4\int Na\, p\, dV_c$$

The static formulas require a regular center, zero surface pressure, vacuum exterior and normalization at infinity. Independent quadrature on the initial star verifies them within 5.5×10^{-14}. The general evolving split is a classical interacting-source counterpart to the earlier positive quantum split: clock progression is an energy derivative in both. It does not identify a classical functional with a noncommuting quantum operator. [R12]



Two distinct realizations of the clock-energy accounting. Left: the exact commuting probe formula at the illustrative fixed lapse α=0.8. Right: the independently re-solved source during active emission, with its clock-linked material contribution and positive complement. The right complement includes momentum, compression, photons and field-response terms; it is not all kinetic energy. The partition residual is 3.13×10^{-14} on this represented state, whose reconstruction offset is 8.77×10^{-7}. No observational parameter is fitted. [R1, §32; R12, §5.]

Emitted radiation carries a continuing clock and energy response

When photons separate from the spherical remnant, their current exterior state still affects its reference-clock normalization. In the vacuum gap:

$$N = \mathcal{C}(T)\sqrt{1 - 2M_{\text{rem}}/r}, \qquad Na = \mathcal{C}(T)$$
$$\mathcal{C} = \exp\left[-\int_{\text{gap}}^{\infty} 4\pi r a^2 (E + P_r)\, dr\right] \tag{113}$$

Holding the exterior canonical photons fixed and varying the inner energy gives the homogeneous version of the same source-constraint variation. For a positive inner energy increment it produces:

$$\frac{\partial M_{\text{complete}}}{\partial M_{\text{rem}}}\bigg|_{F_\gamma} = \mathcal{C}, \qquad \Delta M_{\text{complete}} = \int \mathcal{C}(M_{\text{rem}})\, dM_{\text{rem}} > 0 \tag{114}$$

The factor setting clock normalization is also the factor setting complete-energy response. This is a concrete form of memory in the present emitted-radiation state. A spatially constant C changes inner reference-time pacing; it supplies no extra local radial force in the vacuum gap.

Vacuum separation extends the result to the entire remnant. The exterior radial constraint starts from its inner boundary energy, so the inner state enters through that one energy:

$$H_{\text{complete}} = \Phi(H_{\text{inner}}, F_{\text{ext}}), \qquad \Phi_{H_{\text{inner}}} = \mathcal{C},$$
$$\dot{x} = \mathcal{C}\frac{\partial H_{\text{inner}}}{\partial p}, \qquad \dot{p} = -\mathcal{C}\frac{\partial H_{\text{inner}}}{\partial x}. \tag{115}$$

H_inner includes the inner material, radiation and binding. After the pulse ends, its autonomous canonical variables commute with the separated exterior variables. With no flux across the persistent gap, {H_inner,H_complete}=0. Thus H_inner stays conserved even while exterior photons evolve and C(T) changes. The differential conversion dT_inner=C(T)dT sets the same pacing for every inner Hamilton equation. Complete energy is Φ, generally not the product C H_inner. [R12, §§8-9.]

Writing Ξ=Q²/V for the assigned complete energy of each subsystem gives ∂Ξ_complete/∂Ξ_inner|F_ext=C, because both energies use the same factor c²/(24πG). This is the source-response reciprocity reading of the actual emitted-field state. The corresponding clock burdens combine as

$$u_{\text{total}} = u_{\text{env}} + (1 - u_{\text{env}})u_{\text{rem}}, \qquad u_{\text{env}} = 1 - \mathcal{C}^2, \qquad u_{\text{rem}} = \frac{2GM_{\text{rem}}}{rc^2} \tag{116}$$

This exact composition distinguishes global clock burden from the isolated source reading. The gap's tidal scalar remains $48G^2M_{\text{rem}}{}^2/(c^4r^6)$. Its shorthand $12S^4/(c^4r^4)$ requires the isolated source S, not total reference-clock burden including the external shell. [R12]

Two finite-pulse calculations

The regular model source has $n_{center} = 0.1$, $m_b = 0.8$, $z_0 = 0.2$ and $K = 1$. Solving its support gives $R = 1.0047350068$, $M = 0.12695741727$, $2M/R = 0.2527182121$ and central $N = 0.7203788490$. Two independent radial-mode methods give $\omega_0^2 = 0.1993213746 > 0$ for its unforced initial equilibrium. These are model data and calculated outputs, not astronomical fits.

The engine uses a smooth core window in the initial material label a_0 and a pulse lasting one unit of carried proper age s. The following functions are zero outside $a_0 < R_b$ and $0 \le s < 1$ respectively:

$$\Gamma = 0.01\eta\, w(a_0) g(s), \qquad R_b = 0.8R,$$
$$w(a_0) = \exp\left[-\frac{a_0^2}{R_b^2 - a_0^2}\right], \qquad g(s) = \exp\left[-\frac{s^2}{1-s^2}\right]. \tag{117}$$

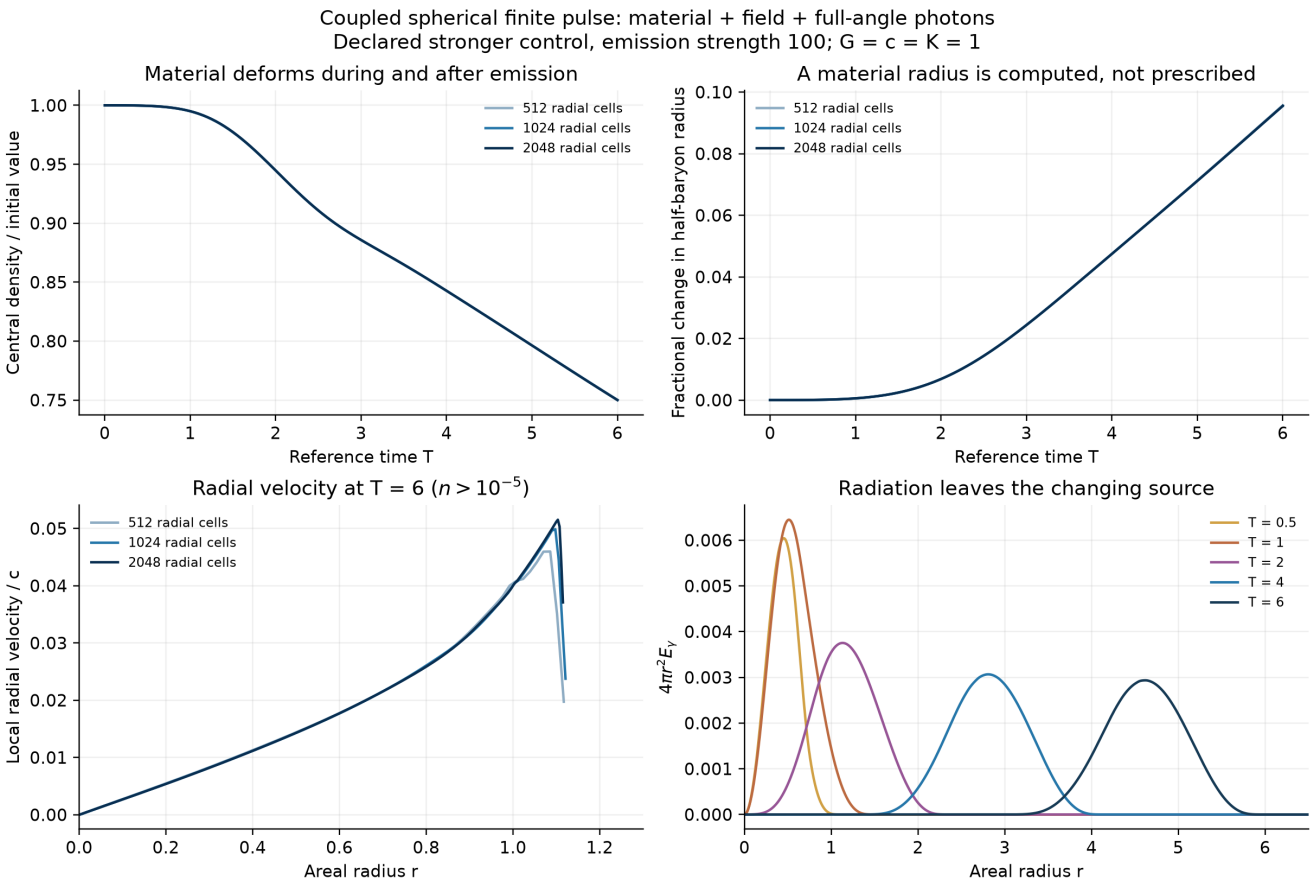
The initial label and proper age are advected with the material. The positive rest-frame angular pattern is $dL/d\Omega = L(1+b\cdot\omega)/(4\pi)$, where ω is the emitted direction. The directed test uses $|b| = 0.75$ and the spherical test uses $b = 0$. These are declared apparatus inputs.

Directed pulse. A positive dipole photon engine lasting one material proper-time unit was propagated through this exact initially bound source. Independent ray integration and the previously calculated material-plus-field inertia give the leading coefficients below. η scales the declared emission strength. [R10]

Quantity	First-order result in the model units
Escaping photon energy	$\eta \times 4.11335348769 \times 10^{-5} + O(\eta^2)$
Escaping axial momentum	$\eta \times 9.75740140357 \times 10^{-6} + O(\eta^2)$
Asymptotic remnant speed / c	$-\eta \times 7.68557018052 \times 10^{-5} + O(\eta^2)$

Ray deflection reduces the coherent directed impulse by 5.115% relative to unchanged directions at the same redshift. The finite-time source/field center projection includes field momentum while photons are nearby; its endpoint is the asymptotic recoil. This result is first order in pulse strength with exact initial binding.

Coupled spherical pulse. Isotropic emission was then evolved with full-angle photons, material deformation and changing gravity. Its net vector recoil is zero by symmetry, while internal radial motion is nonzero. The separately labelled $\eta = 100$ control lowers central density by about one quarter by $T = 6$. Its predicted initial central response is one seventh of the frozen-field response, demonstrating the effect of the coupled field before the full pulse is integrated. [R12]



Computed material and photon histories for the declared stronger spherical control. The field is re-solved throughout. Radial resolution is compared without fitting a response curve. The velocity panel shows cells with n > 10^{-5}; full-source diagnostics retain all material, including the more difficult surface region.

Endpoint energy comparison	G = c = K = 1
Positive internal radial-motion energy	$3.14660481 \times 10^{-5}$
Quadratic estimate	$3.14454727 \times 10^{-5}$
Deformation above same-composition equilibrium	$7.05082197 \times 10^{-6}$
Total excitation	$3.85168701 \times 10^{-5}$
Motion increment including the fixed exterior photons	$3.14173632 \times 10^{-5}$

These numbers use 2048 radial cells and 32 uniform angular cells. The last two radial levels differ by 0.0574% in intrinsic motion energy; a separate matched angular refinement changes it by 0.00204%. The independent angular field-equation residual is 0.2882% of its peak changing-source scale, with a finite refinement floor. The reconstructed remnant mass differs from its raw grid value by 3.40×10^{-7}, about 1.1% of the small motion increment. The much smaller quadrature residual of the exact energy identity is not the precision of the evolving state.

The common-energy theorem was also tested during active emission. Independent nonlinear constraint re-solves agree with the rest-energy, momentum and photon derivatives to relative 7.55×10^{-11}, 7.22×10^{-8} and 3.09×10^{-11} on that represented state. Its baseline reconstruction offset is 8.77×10^{-7}. A separate finite material displacement verifies the force derivative, with centered errors decreasing quadratically.

The complete result is therefore more than an energy rearrangement: regular evolving sources realize the same clocks, force, fuel exchange, light transport and positive internal motion energy. The directed first-order response and nonlinear spherical deformation are complementary achievements. The spatially resolved nonspherical finite-amplitude recoil remains a further computation, using the already specified local source and field equations. [R9-R12]

7. The same source energy fixes a stationary photon transfer

The complete source derivative gives a direct optical consequence. Consider neighboring stationary states of the two supported shells introduced earlier. Let the inner shell lose a small proper energy δ and the outer shell receive ε. If there is no additional independent work and the complete energy is unchanged,

$$0 = d\overline{\mathcal{H}} = -\alpha_1\delta + \alpha_2\varepsilon, \qquad \frac{\varepsilon}{\delta} = \frac{\alpha_1}{\alpha_2} \tag{118}$$

Independently, the retained stationary photon Hamiltonian conserves αE_γ,local and gives the same ratio. Supplying the Planck-Einstein relation E_γ=hν makes it the frequency ratio as well. The material source accounting and the probe propagation agree.

For the worked example take U_support,i=k_i(R_i-a_i)²/2. In G=c=1 units with $a_1=1$, retain $(b_1,b_2)=(0.08,0.12)$, $(k_1,k_2)=(0.096,0.02304)$ and $(a_1,a_2)=(1,2.5)$. The stationary radii are outputs: $(R_1,R_2)=(0.9605095316,2.3569337553)$. They give

Worked stationary source	Calculated value
Inner source clock α_1	0.8682710811
Outer source clock α_2	0.9158803457
Received/emitted proper energy	0.9480180300

These clocks come from the solved elastic-shell source, with fixed material parameters and radially stable supporting equilibria. The ratio is not fitted. The same complete energy supplies the first finite correction. Define r_α=α₁/α₂ and C_ij=∂²H/(∂b_i∂b_j), including support relaxation. Then

$$\varepsilon = r_\alpha\delta - \frac{C_{11} - 2r_\alpha C_{12} + r_\alpha^2 C_{22}}{2\alpha_2}\delta^2 + O(\delta^3) \tag{119}$$

For the stated source, the quadratic coefficient is 0.3992779816 in G=c=1 units. Fixing the radii instead gives 0.3411942744: it is a different supported process. This finite relation compares

neighboring stationary complete states. A real finite transit may also contain recoil, excitation and radiation, which the finite-pulse calculations above keep explicitly. [R1, §§34-35; R10-R12.]

Thermal equilibrium is a related common-energy consequence

Supply the ordinary local entropy relation ∂s_i/∂e_i=1/Θ_i. Extremizing total entropy at fixed complete H then gives 1/Θ_i=λβ_i, hence

$$\beta_i \Theta_i = \Theta_\infty \tag{120}$$

For held sources β=α. Their emitted thermal spectra arrive at one observer with the same temperature Θ_o=Θ_∞/α_o. A varying burden does not by itself create a thermal spectral distortion. The common source energy and the source-clock conversion account for the cancellation. The entropy principle is the additional thermodynamic input. [R1, §43.]

8. Source-volume evolution connects massive cooling and photon redshift

The homogeneous optical branch retains a common spatial scale a(T), lapse α(T) and a fixed-baryon material domain V_D=V_D0 a³. Its canonical probe Hamiltonian is

$$\mathcal{H} = \alpha E, \qquad E = \sqrt{E_0^2 + c^2 |\mathbf{p}|^2 / a^2} \tag{121}$$

Here E is local normal-observer energy, not the stationary reference energy used in the orbital chapter. E_ref below denotes that local energy at the chosen reference epoch. Homogeneous translation symmetry conserves p_i. The fixed-mass source law gives Q_b²=Q_b0²a³. Eliminating a yields

$$\begin{gathered} E^2 - E_0^2 = (E_{\mathrm{ref}}^2 - E_0^2)(Q_b/Q_{b0})^{-4/3}, \\ (E^2 - E_0^2)^{3/2} Q_b^2 = \text{constant}. \end{gathered} \tag{122}$$

One relation covers both massive motion and light. In the photon limit E₀=0 it becomes E_γ³Q_b²=constant. In the slow massive limit, K=E-E₀ scales as Q_b$^{-4/3}$. The clock derivative is still H_E₀=αE₀/E=β, and the group velocity returns the original capacity budget.

$$E_\gamma^3 Q_b^2 = \text{constant}, \qquad \frac{E_o}{E_e} = \left(\frac{Q_{b,e}}{Q_{b,o}} \right)^{2/3} \tag{123}$$

Retain the source pace when differentiating in time

If the branch additionally retains dV_D/dT=Q_b, then a_T/a=Q_b/(3V_D). Differentiating the same energy gives

$$\frac{dK}{dT} = -\frac{Q_b}{3V_D}\frac{K(K+2E_0)}{K+E_0}, \qquad \frac{d\ln(\beta/\alpha)}{d\ln Q_b} = \frac{2}{3}\left[1-(\beta/\alpha)^2\right]. \tag{124}$$

Cooling translation increases the share of available capacity assigned to internal progression. The absolute β also contains the evolving burden through α. The source law, the retained homogeneous motion and the stated volume pace have different roles; none is replaced by a fitted attenuation coefficient. [R1, §§14-15.]

Moving boundaries require the mass-compensated identity

A front that sweeps up new material has a changing included mass M_h. M_h is the baryonic material mass included by the advancing front, and ρ is that material's homogeneous density. Baryon conservation fixes ρ∝a⁻³, while the included M_h changes. This is the material source reading, not a complete energy containing an additional radiation component. At the same density, the source law gives (Q_h/M_h)²=24πG/ρ. Since E_γ∝a⁻¹, the correct moving-domain relation is

$$E_\gamma^3\left(\frac{Q_h}{M_h}\right)^2 = \text{constant} \tag{125}$$

Using E_γ³Q_h² with changing M_h would mistake a changing inclusion domain for a change in photon energy. Similarly, a held normalized rate Q_held=Q_b/α enters the fixed-material relation as E_γ³α²Q_held²=constant. The clock conversion must travel with the source label. [R1, §16.]

Phase, arrival intervals and brightness

Use conformal time dζ=αdT/a. A massless phase Φ=k·x-c|k|ζ has reference frequency ω_T=αc|k|/a and local frequency ω_local=c|k|/a. The observer divides by the same α that appears in the reference phase rate. The lapse remains important in travel time; it is not a second local redshift factor.

$$\chi_{eo} = \int_{T_e}^{T_o}\frac{c\alpha(T)}{a(T)}\,dT, \qquad Z = \frac{a_o}{a_e}, \qquad \frac{dt_o}{dt_e} = \frac{E_e}{E_o} = Z, \qquad D_L = ZD_G, \qquad F_o = \frac{L_e}{4\pi D_G^2 Z^2}. \tag{126}$$

D_G² is the received beam area divided by emitted solid angle; in the flat homogeneous reading D_G=a_oχ_eo. Photon-number conservation and the ray geometry are the premises of the brightness relation. Redshift reduces energy per photon and stretches arrival intervals. If delayed propagation is already included in the integral, multiplying by another exposure factor would count that part twice. [R1, §17.]

Directional response is information that a volume alone cannot carry

For a supplied homogeneous anisotropic reading with directional rates H_i, let H be their mean and σ_ij their trace-free part. Actual rays have direction n_i. The same local Hamiltonian gives

$$\frac{d\ln E_\gamma}{dt} = -H - \sigma_{ij}n^i n^j, \qquad \frac{d}{dt}\ln(E_\gamma^3 Q_b^2) = -3\sigma_{ij}n^i n^j \tag{127}$$

The simple cubic law is the shear-free limit. This is useful for SET because source volume, field organization and photon direction now have explicit places in one accounting. A volume history fixes a part of the optical response; the remaining directional terms require the source's actual spatial evolution. [R1, §§42-45.]

A source-volume identity can fix a spectral moment

One exact spectral control makes this last distinction measurable in principle. Supply collisionless propagation from an initially uniform blackbody temperature Θ_e through homogeneous relative directional scale factors a_i between emission and reception. A ray emitted along n_e has energy ratio f=[Σ_i n_e,i²/a_i²]$^{1/2}$. The angular map is dΩ_o=f^{-3}dΩ_e/det(a_i). With ⟨f⟩_o=(4π)$^{-1}$∫f dΩ_o, its observed third temperature moment has the exact Jacobian cancellation

$$\langle\Theta_o^3\rangle_o = \frac{\Theta_e^3}{\det(a_i)}, \qquad \Theta_n^3 Q_{b,o}^2 = \Theta_e^3 Q_{b,e}^2, \qquad \Theta_n^3 = \langle\Theta_o^3\rangle_o \tag{128}$$

The source law fixes this photon-number temperature moment. It does not make all directions have the same temperature. Positive normalized observer-direction weights and convexity give a separate bolometric spectral test:

$$\mathcal{D}_{\mathrm{mix}} = \frac{\langle\Theta_o^4\rangle}{\langle\Theta_o^3\rangle^{4/3}} - 1 \geq 0 \tag{129}$$

Equality requires one received temperature. Directional source organization can therefore preserve the source-volume identity while changing the received spectrum. The supplied collisionless radiation and angular weighting are part of this control; it is not a new fitted cosmological history. [R1, §§42-43.]

What this supplies for the luminosity program
The research supplies exact source/energy/clock identities and a propagation rule with the clock factors kept consistently. Selecting a complete cosmological source history still requires that the same interacting source evolve with that history and produce those rays. The identities above are necessary links for that task; they do not turn an inverse reconstruction of the required work into a solved interacting source.

9. What should remain active in the SET research program

The most consequential result is the **complete-energy variation including its force derivative**. The source does not merely carry a clock factor assigned from an external background. Its own field is re-solved, and the same complete energy then supplies clock progression, material velocity, momentum exchange and photon propagation. The finite-pulse calculation places those derivatives on actual interacting source states.

The independent Q-boundary test adds a different achievement. Candidate supported sources can satisfy their own static field and force equations and their clock-current energy comparison while missing the source-law boundary. The source requirement therefore has genuine discriminating content in the tested class. The source test and the evolving common-energy theorem should be retained together, with their premises visible.

Retained result	Why the next calculation needs it
Complete source rather than isolated field energy	The actuator, material stress, binding and escaping radiation all contribute to energy and momentum.
Source-law boundary test	It distinguishes candidate source responses without using an observed orbit to choose the coefficient.
Canonical clock/velocity/force derivatives	They turn consistent accounting into an actual local motion law.
Positive quantum and classical complements	They reconcile internal progression with other energy contributions without mistaking that complement for ordinary kinetic energy.
Electromagnetic curl, constraints and turning	They identify a concrete common-capacity vector realization and the material current it needs.
Positive finite radial-motion energy	It includes the field's readjustment and supplies a strong control for nonlinear inertia calculations.
Source-clock photon energy relation	It connects complete source exchange, optical redshift and the actual clock normalization.
Domain, orientation and averaging bookkeeping	It prevents boundary changes, hidden fluctuations or omitted directional work from being read as new source physics.

The next decisive calculation is now specific

The finite spherical pulse has a zero net vector recoil but a nonzero internal response. The directed first-order calculation supplies a recoil benchmark. The next full test joins those two achievements: evolve a regular source through an axisymmetric directed pulse, keep the actuator and photons, and compare the remnant's final complete momentum with the same energy-generated force law. The source equation, coupling and impulse prescription should remain unchanged between the benchmark and the finite run.

For electromagnetism and quantum motion, the strongest continuation is similarly concrete: use the same energy to derive the material current and its field response, then apply an independent complete-source boundary test where it is available. Equal units, a positive square or a successful clock conversion alone should not be asked to select a signed charge or a matter representation.

For cosmology, the retained optical identities specify the work and ray accounting that an interacting source must realize. A new source construction can be judged against them without restarting the algebra. These are three applications of the same research discipline: keep the complete state, vary its common energy, and test its source reading independently.

10. Three information checks that must survive future simplification

Averaging cannot erase unresolved motion

Average individual capacity relations at one common burden and in one common spatial/clock reading, using positive normalized weights. Separating means and variances gives the exact identity

$$S^2 + |\langle \mathbf{V} \rangle|^2 + c^2 \langle \beta \rangle^2 = c^2 - \mathrm{Var}(\mathbf{V}) - c^2 \mathrm{Var}(\beta) \tag{130}$$

Equal counterpropagating streams have zero mean translation but nonzero motion and energy. The same distinction matters for quantum standing waves and a spherically deforming source. A vanishing center-of-mass velocity does not make internal motion vanish. [R1, §§22-23, 36; R12.]

A squared burden does not retain a front's orientation

In the retained homogeneous null-front prescription, the signed variable y obeys the following equation, with $N_a = \ln a$. Here u is the burden fraction and $q = \alpha^{-2}$ is the expansion-side energy ratio of that prescription:

$$\frac{dy}{dN_a} = \frac{1+y^2}{2}(\sqrt{6} - y), \qquad u = \frac{y^2}{1+y^2}, \qquad q = 1 + y^2,$$
$$\frac{d\ln q}{dN_a} = \sqrt{6}\, y - y^2. \tag{131}$$

Separating the first equation gives $\ln(1+z) = F(y_0) - F(y)$, with

$$F(y) = \frac{\ln(1+y^2) + 2\sqrt{6}\arctan y - 2\ln|\sqrt{6} - y|}{7}. \tag{132}$$

The same u and q accompany two possible signs of y and therefore two work rates. The original outward-front history chooses $y \geq 0$. At the retained reception value $y_0 = 1.7789876897$, its zero-radius launch $y = 0$ gives $z = \exp[F(y_0) - F(0)] - 1 \approx 2.724443$ at finite density and finite scale factor. It has $u = 0$ and $\alpha = 1$, not a saturated-capacity photon extinction. A signed earlier continuation changes the front preparation and reaches a different, singular endpoint. This is a branch-domain result, not a universal SET observable horizon. [R1, §§18-20.]

An actual-source audit also supplied a useful exact exclusion. For the tested initially uniform pressureless spherical dust class, $m=\kappa r^3$ and the shell equations imply $V^{(5)}=2\kappa V^{(3)}$. After the lower target derivatives are matched, the fifth volume derivative is 9/2 rather than 153/4 in the stated normalized units. The same unaltered dust state cannot sustain the prescribed front/energy history. This result tells the next source calculation where interaction stress or a different admitted source class becomes essential; it does not reject all SET cosmological states. [R1, §24.]

A tidal magnitude and a clock burden need their solved relation

For the retained rotating exterior put $m=GM/c^2$, $a_s=J_s/(Mc)$, and $\Sigma_r=r^2+a_s^2\cos^2\theta$. The quadratic curvature contractions K and its dual P combine, for one orientation convention, as

$$K+iP=\frac{48m^2}{(r-ia_s\cos\theta)^6} =\frac{3Q_s^4}{64\pi^4c^4R_0^6(r-ia_s\cos\theta)^6}, \qquad \sqrt{K^2+P^2}=\frac{48m^2}{\Sigma_r^3}. \tag{133}$$

The source radius R_0 and the measurement coordinate r retain separate roles. K alone can change sign in a rotating exterior, while the displayed magnitude stays nonnegative. Neither has the units of S. The expression is an exact Q conversion of the retained rotating field, not a new derivation of that field. Its practical value is to prevent a neutral-spherical shortcut from being promoted to a universal energy or capacity density. [R1, §13.]

11. Verification, notation and retained premises

The consolidation includes new independent symbolic checks of the source conversion, source composition, two-source clock derivatives, orbital scaling, perihelion coefficient, source-volume cooling and the front primitive. The quantum audit separately repeats noncommuting positive-operator and spin-capacity controls. The evolving-source chapter preserves the original nonlinear re-solve and field-evolution checks. Fresh independent programs also re-solved a mixed matter-photon constraint and the supported capacitor alternatives without importing the original solvers. They recovered the clock, momentum, photon and position derivatives, the positive finite motion increment, and the withheld Q discrimination. Algebraic residuals and evolution errors describe different tests; neither is an observational uncertainty.

Layer	What is supplied or retained	What follows in this document
Capacity and finite-domain source law	The invariant, domain convention and source universality	Equivalent source readings, quadratic exchange and correct normalizations.
Stationary source realization	The selected two-response field energy, material stress and complete support boundaries	Charged solution, source clocks and an independent response-coupling test.

Layer	What is supplied or retained	What follows in this document
Local common evolution	Canonical field structure, common lapse/shift, declared local vector class	Curl structure, Gauss constraint, common wave speed and Maxwell branch within that class.
Quantum realization	A specified wave operator, domain, charge/spin structure where stated	Bounded clock derivative, positive split, source-clock momentum and within-class coupling restrictions.
Evolving regular source	Spherical common field equations, canonical fluid/photons, causal material law and stated fuel pulse	Actual energy/force derivatives, consistent exchange, positive motion energy and finite deformation.
Homogeneous optics	Common probe propagation, source domain, homogeneity; source pace when used	Cooling/redshift identities and brightness accounting for the supplied history.

These premises are not independent adjustable coefficients in every numerical example. They identify the retained theory realization and the matter experiment being tested. The numerical comparisons hold them fixed within each declared test.

Symbol or reading	Definition and sector
Q_s, Q_b, Q_h	Complete source label; fixed-baryon material label; moving-front label. Each carries its stated volume and mass.
Q_m or Q_clock	The specified reference rate divided by a material clock factor; not a second source mass.
α or N; β	Held/normal clock lapse; moving material clock progression. In the radial source β=N/W.
V_ref, dV, dV_c	Specified source-reference volume; physical proper volume; coordinate volume. In radial areal coordinates dV=a dV_c.
μ	GM_s in the orbital chapter; rest-mass-and-fuel energy per baryon in the inertia chapter, where G=c=1.
W; h	Lorentz factor and specific enthalpy in the material chapter. Planck's h is used only in the photon-frequency relation.
Γ; dΓ	Local fuel-emission rate; separately, the explicitly defined canonical photon phase-space measure.
K_0, K_w, K	Quadratic motion energy in the capacity chapter; wave energy-squared operator in the quantum chapter; polytropic coefficient or explicitly named tidal contraction in their respective chapters.
E, H, M	Local probe energy or explicitly named stationary energy; complete or probe Hamiltonian; source mass, also source energy when c=1.
e, ρ_e, j_e	Signed electric charge, charge density and current. They are not gravitational source Q.

12. Research record and retention ledger

The scope begins with the source-history, quantum-phase and capacity research record and ends with the nonlinear inertia continuation. Earlier source-selection and integrated papers supply the retained starting framework. The records below are preserved unchanged in the research workspace; their named sections and accompanying check scripts provide the detailed calculations.

[R1] SET source history, quantum phase and capacity research

The 69-page starting report. Retained sections: source/orbit first laws (§§1-5); complete charged sources and response selection (§§6-9); rotating source/area/tidal accounting (§§10-13,39); source-photon transport (§§14-17,34-35,40,43-45); quantum positive split (§§25,30,32); domain and averaging controls (§§18-24,36,42).

Workspace record: outputs/set-front-optics-research-2026-11-05/RESEARCH-REPORT.txt

[R2] SET common capacity and electromagnetic evolution research

Canonical curl/Gauss/common-speed selection within its declared vector and derivative class.

Workspace record: outputs/set-em-dynamics-selection-2026-11-05/coordinator/RESEARCH-REPORT-TEXT.txt

[R3] SET source energy, field content and electromagnetic selection

Representation audit, alternative supported matter, exact-cone restriction, work versus turning, charged probe source-clock reading.

Workspace record: outputs/set-em-representation-and-class-selection-2026-11-05/coordinator/RESEARCH-REPORT-TEXT.txt

[R4] SET quantum source, spin and electromagnetic bridge

Kinetic commutator, source-clock momentum, Pauli current, spectrum and distinct microscopic/source normalizations.

Workspace record: outputs/set-qm-em-bridge-2026-11-05/RESEARCH-RECORD.txt

[R5] SET: selecting charged phase and spin coupling

Positive spin-capacity family, affine clock restriction, field-dependent dipole causality test and surviving constant-coupling controls.

Workspace record: outputs/set-phase-spin-selection-2026-11-05/RESEARCH-REPORT.txt

[R6] SET: inertia and reorganization of an established source field

Historical scalar-memory acceleration control. Retain its complete-work lesson; the later selected-field calculation is the active inertia branch.

Workspace record: outputs/set-inertia-source-reorganization-2026-11-05/RESEARCH-REPORT.txt

[R7] SET selected fields: a conserved acceleration test and its inertial response

Weak-field actuator, support/current terms, reaction and radiated energy accounting.

Workspace record: outputs/set-selected-field-impulse-2026-11-05/RESEARCH-REPORT.txt

[R8] SET nonlinear acceleration, source impulse and complete inertia

Boundary-charge inertia and exact radiation-driven exterior benchmark.

Workspace record: outputs/set-nonlinear-inertia-2026-11-05/RESEARCH-REPORT.txt

[R9] SET: a regular supported source, a specified impulse, and calculated inertia

Regular source and exact-binding slow-translation inertia; input source for later pulses.

Workspace record: outputs/set-regular-source-impulse-2026-11-05/RESEARCH-REPORT.txt

[R10] SET finite-pulse response

First-order directed pulse, actual escaping photon energy/momentum and remnant recoil benchmark.

Workspace record: outputs/set-finite-pulse-response-2026-11-05/RESEARCH-REPORT.txt

[R11] SET: toward the full nonlinear finite-pulse response

Nonlinear transport, source and boundary machinery. The subsequent record supplies the accepted spherical finite run.

Workspace record: outputs/set-full-pulse-evolution-2026-11-05/RESEARCH-REPORT.txt

[R12] SET inertia, deformation and complete-source energy

Evolving source-energy/force theorem; exact transfer; positive finite motion and clock-energy split; exterior photon memory; audited finite spherical pulse.

Workspace record: outputs/set-inertia-two-hour-2026-11-05/RESEARCH-REPORT.txt

Supporting results preserved in the library

The gravitational-atom comparison, binary first-law scaling, rotating scattering controls, spectral mixing, Fierz identities, quantum coherence bound and source-support Hessians remain useful specialized tools. Their premises and derivations are retained in R1 and the synthesis audit notes. They are not all repeated here because the main chain does not require every specialized example.

The historical scalar-memory kernel, failing long-duration numerical controls and branch continuations that change the optical history remain research evidence. They are not substituted for the accepted dynamical calculation. The full records preserve what each control tested and what it did not establish.

Mathematical context for the retained structures

The orbital action/mean-clock identities have a standard counterpart in A. Le Tiec, A Note on Celestial Mechanics in Kerr Spacetime (2014). The source-law form in this synthesis keeps the source radius, source charge and physical spin derivatives explicit.

For the retained material Hamiltonian and its canonical labels, see J. D. Brown, Action functionals for relativistic perfect fluids (1993). For the spherical collisionless-field reduction, see C. Gundlach, The Einstein-Vlasov system in spherical symmetry (2016). These references identify the established structures used by the retained completion; the displayed SET derivations supply their source, clock and complete-energy reconciliation.